

Evaluating the Effectiveness of Open-Source LLMs for Multi-Domain Bengali Text Classification with Zero-Shot and Few-Shot Prompting

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ABSTRACT Large Language Models (LLMs) have advanced multilingual NLP significantly, yet their effectiveness for Bengali, a globally important but low-resource language, remains understudied due to limited linguistic resources and the absence of a standardized evaluation benchmark. This is due to the unavailability of large linguistic resources and the lack of a standard benchmarking framework, which has yet to be established. This study aims to address the underrepresentation of Bengali text classification by leading a systematic evaluation. It involves assessing multiple open-access LLMs on Bengali text classification tasks and observing their limitations and regular error patterns. The procedure consists of six open-source LLMs, which include LLaMA-3.2-3B, Mistral-V3-7B, DeepSeek-R1-8B, Phi-4-14B, Gemma-2-27B, and Qwen-2.5-72B, and are evaluated using four critical text classification tasks in Bengali, namely sentiment analysis, emotion recognition, hate-speech detection, and fake-news classification. Furthermore, two prompting methods are compared in the context of zero-shot and few-shot prompting using the vLLM model to investigate the adaptability of the model in resource-limited settings. The outcomes of our study indicate that, regardless of the dataset, the bigger multilingual models always performed better in comparison with the smaller ones. When training in zero-shot, Qwen-2.5-72B showed the best performance at 79.9% accuracy and 79.88% F1-score, showing amazing accuracy on hate-speech detection. In the case of fake news, Gemma-2-27B performed best at 84.45% accuracy and 84.33% F1-score. With a few-shot set-up, Gemma-2-27B and Qwen-2.5-72B demonstrated a high level of performance in terms of F1-score, 84.17 and 65.86, respectively. After the final observation, the study reveals that the model size, quality of multilingual pretraining, and contextual samples have a vital impact in low-resource settings.

INDEX TERMS Large Language Models (LLMs), Bengali NLP, Zero-shot prompting, Few-shot learning, Text classification, Open-source, Multilingual models, Natural Language Processing (NLP), Low-resource languages.

I. INTRODUCTION

Large Language Models (LLMs) have revolutionized natural language processing. Large Language Models (LLMs) facilitate enhanced efficiency on different types of tasks, without large-scale training related to the task. The models utilize context-based learning to comprehend the pattern from the given prompts, which in turn decreases the requirement for a large labelled dataset. Nevertheless, it is notable that most achievements of LLMs are observed in high-resource languages as English, while the low-resource

languages like Bengali do not receive enough attention. Bengali is a rich language with immense diversity, even though it is considered to be the 7th most spoken language in the world, and there are over 230 million Bengali speakers; this widely spoken language is still underrepresented [45]. In [10], [29], authors mention that exclusive-use licensed models like GPT-4 and Claude-3.5 present moderate capabilities in processing the Bengali language. On the other hand, publicly accessible models like Gemma-2 and LLaMA-3.1 show disappointing outcomes in the same task.

The main drawbacks are data quality and the substandard performance of well-known robust LLMs. The majority of the Bengali language processing conducted previously depends heavily on the machine-generated translated datasets like Bangla-Alpaca or Bangla-Orca translated version of the original Stanford Alpaca and OpenOrca dataset [48], hence it causes a rise in various errors and poor performance from the models [28]. Moreover, an extensive review shows that well-known robust LLMs encounter difficulties just to reach satisfactory outcomes in Bengali tasks. BenLLM-Eval finds that from time to time, GPT-3.5 and Claude models' performances match the performance of fine-tuned Bengali models on a few tasks [3]. Furthermore, in the case of zero-shot, the performance of the majority of LLMs are substantially inadequate, and publicly accessible models (like LLaMA-2-13B) perform worse.

In case of accuracy, sizeable differences are observed between English and Bengali text-related tasks, which is more significant for smaller model series (e.g., Mistral) [6]. This difference in accuracy can be overcome via in-context based learning. Cahyawijaya *et al.* [43] shares that by presenting a few suitable samples, the LLMs showcases significantly enhanced understanding of the low-resource languages. Nonetheless, the research comprising Bengali language-related LLMs are still scarce.

Recently, researchers have taken the initiative to resolve the accuracy difference that LLMs face by building exclusive Bengali language resources. Nahin *et al.* [21] presented the first substantial pretrained Bengali LLMs named TituLLMs. The authors also introduced five new Bengali baseline datasets and showed that the performance of Bengali-specific models is far better than other language models for native tasks. TigerLLM introduced in the study [10], trained on a meticulously refined Bengali dataset, showcases a new benchmark for Bengali language processing, surpassing the previous publicly accessible models. GPT-3.5 Turbo achieved more than 97% accuracy in Bengali hate speech detection by utilizing few-shot prompting, which outperforms streamlined transformer models [22]. Regardless of the mentioned accomplishments, there is still a limited number of extensive multi-domain baseline and systematic prompt assessments. The majority of previous research focuses on individual tasks or utilizes translated English baselines, which led to a rise in noise [6], [28]. Confronting these differences in outcomes is vital for creating a multilingual natural language processing model. The main objective of our research is to provide a systematic evaluation of publicly accessible LLMs based on four Bengali text classification tasks under zero-shot and few-shot prompting conditions.

A. BACKGROUND AND RESEARCH GAP

According to a few recent studies, model dimension and the scope of the training dataset have an immense influence on the downgrade of performance [47]. In a study [35], bigger models show amazing performance in zero-shot

tasks for high-resource languages like English; however, the performance plummets for low-resource languages. And many LLMs can learn and solve problems with the help of few-shot prompting [46]. Cahyawijaya *et al.* [43] reveals that introducing a few samples via few-shot prompting helps enhance the model performance across 25 undervalued languages. However, the Bengali language is not specifically included in this list of undervalued languages. Because of the lack of large-scale pre-trained LLMs focused on the Bengali natural language processing domain, most advancements rely heavily on streamlined models like BanglaBERT [4]. During the research, Bhowmik *et al.* [6] found that current multilingual datasets consist of around 30 billion Bengali text tokens, in contrast English language consists of a trillion tokens, which results in poor model quality. In [3], Kabir *et al.* build BenLLM-Eval, a baseline model capable of summarization, classification, and QA tasks in the Bengali language. Additionally, they notice that even though GPT-3.5 sometimes outperforms the fine-tuned models, its performance is poor for a zero-shot system. Likewise, Nahin *et al.* [21] introduced TituLLMs and native baselines, showcasing enhanced performances across multilingual models. Regardless of these initiatives, significant research gaps are still present. Bengali language classification has not been analyzed appropriately via zero-shot and few-shot settings. Latest publicly accessible models, namely Qwen, Mistral, or Phi-4, have yet to be evaluated on Bengali text classification tasks, and overall error assessment of the models is lacking. To tackle these research gaps, our study aims to provide a systematic evaluation of six publicly accessible LLMs across four Bengali text classification tasks with further insight into their limitations and error patterns.

B. RESEARCH QUESTIONS

To analyze the performance of publicly accessible LLMs for Bengali text classification tasks, this study is guided by the following research questions:

- **RQ1:** What is the performance of publicly accessible LLMs on Bengali text classification over several different domains, consisting of sentiment analysis, emotion recognition, hate speech detection, and fake news detection?
- **RQ2:** What is the impact of model size and pretraining of multilingual models on text classification performance, specifically for low-resource languages like Bengali?
- **RQ3:** Is there any noteworthy improvement noticed in accuracy, recall, and F1-score when utilizing few-shot prompting, in contrast to zero-shot prompting, in the Bengali text classification task?
- **RQ4:** What sort of incorrect classification, limitations, or systemic errors are usually found during LLMs performance for Bengali text tasks, and what type of limitations are stipulated by these error patterns?
- **RQ5:** How uniform are various publicly accessible LLMs in classifying diverse Bengali natural language

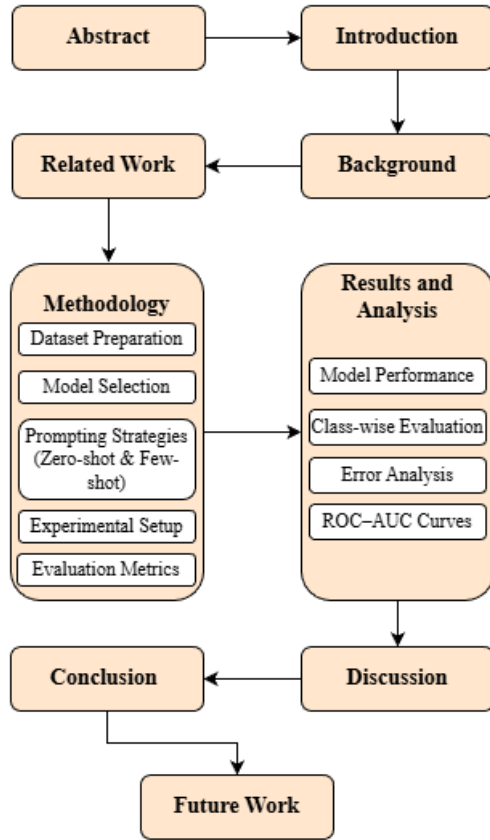


FIGURE 1: Paper organization.

processing tasks, when assessed using a unified, prompt-based pipeline?

C. MAIN CONTRIBUTIONS

This research presents several noteworthy contributions to the rising field of the Bengali natural language processing domain:

- 1) **Comprehensive Evaluation:** We perform a comprehensive evaluation of six latest publicly accessible LLMs: LLaMA-3.2-3B, Mistral-V3-7B, DeepSeek-R1-8B, Phi-4-14B, Gemma-2-27B, and Qwen-2.5-72B, on four types of Bengali text classification tasks: sentiment analysis, emotion recognition, hate speech detection, and fake news classification.
- 2) **Unified Prompting Framework:** We apply a systematic evaluation pipeline utilizing the vLLM structure, while using zero-shot and few-shot prompting to maintain consistency and reproducibility over the models' experiments.
- 3) **Performance Insights:** The overall outcome highlights the significantly better performance of the larger multilingual models (like Qwen-2.5-72B, Gemma-2-27B), specifically in complicated tasks (like emotion recognition), in contrast to smaller

models. This shows the significance of model scale and multilingual model training.

- 4) **In-Context Learning Benefits:** We realized that few-shot prompting increases average recall by over 6%, exhibiting that context-based learning samples improve the model's understanding of low-resource languages like Bengali.
- 5) **Detailed Error and Class-Wise Analysis:** We utilized comprehensive evaluation metrics (Accuracy, Precision, Recall, F1-score) and confusion matrices. This enables recognizing common model errors and recommending an approach for future improvements.

D. PAPER STRUCTURE

The paper is organized into the following sections: First, Section II presents a literature review of prior research on Large Language Models (LLMs), multilingual natural language processing, and Bengali text classification. Section III provides a detailed explanation of the chosen models, dataset preparation, experiment structure, and application of zero-shot and few-shot prompting. Section IV reports the overall performances of the models using evaluation metrics for four types of Bengali text classification tasks, and also presents a comparison among the chosen models. Section V provides detailed insights into the performance patterns, overall findings, common mistakes, and errors related to misclassification. Lastly, Section VI summarizes the contribution of the research, notes the limitations, and recommends future direction of work.

II. RELATED WORK

Large Language Models (LLMs) for the Bengali language are currently gaining growth, similar to previous advances in Natural Language Processing. Though Bangla is one of the most widely spoken languages in the world, it still lags behind in the field of computational linguistics due to limited annotated corpora, ineffective tokenization, and inadequate domain coverage. In recent times, the focus has been on model adaptation, resource generation, and zero- or few-shot prompting strategies to improve Bengali text understanding and classification across multiple domains.

A. PROGRESS IN BENGALI LLMS AND BENCHMARK DEVELOPMENT

For Bangla text classification, Small open source models like Gemma-2B, TinyLLaMA, Falcon are evaluated by Dehan *et al.* [1]. TinyLLaM model outperformed here for detecting fake news, whereas BanglaBERT showed its excellence for emotion recognition. By fine-tuning RoBERTa and BERT on Bengali movie reviews, Rahman *et al.* [2] achieved 88% accuracy, confirming that transformer-based models are more effective than traditional machine learning classifiers. Kabir *et al.* [3] proposed BenLLM-Eval, a comparison platform for evaluating models like GPT-3.5, Claude-2, and LLaMA-2 across a variety of Bengali NLP tasks. Their research showed that fine-tuned Bengali models outperform others

under zero-shot evaluation settings. Bhattacharjee *et al.* [4] proposed BanglaBERT, a Bengali-specific language model trained on 27 GB of native text, which demonstrated strong sentiment analysis and named-entity recognition (NER), establishing a foundation for further research. Mahmud and Mahmud *et al.* [5] created a hybrid sentiment model that combined lexicon-based scoring and BanglaBERT embeddings, resulting in a 93% F1-score and highlighting the benefits of mixing symbolic reasoning with deep neural representations.

Bhowmik *et al.* [6] assessed multilingual LLMs, including Qwen, Mistral, and DeepSeek, on Bengali benchmarks and found that tokenization and translation noise were the primary obstacles to limiting accuracy. Al Nazi *et al.* [7] evaluated open-source and closed-source models, including GPT-4o, Gemini, and BLOOM, in zero-shot and few-shot situations, emphasizing the significance of rapid development in model performance. Prome *et al.* [8] used metaphor-based prompting for hate speech identification, which enhanced interpretability and fairness. Banda *et al.* [9] investigated few-shot learning for multilingual agricultural question answering and shown that LLMs may generalize efficiently with minimal supervision. Raihan and Zampieri *et al.* [10] released TigerLLM, a family of open-source Bengali models trained on the Bangla-TextBook and Bangla-Instruct corpora with performance comparable to GPT-3.5. Mahfuz *et al.* [11] highlighted the continued need of Bengali-specific models, stating that multilingual systems frequently fail to capture language-specific syntax and semantic nuances, highlighting the importance of cultural and linguistic alignment during pre-training.

B. RESOURCE CREATION AND CROSS-DOMAIN STUDIES

Mahtab *et al.* [12] developed BanNERD, a comprehensive 29-domain Bengali NER dataset intended to improve contextual generalization in named-entity recognition algorithms. In a similar vein, Shafayat *et al.* [13] published BEQA, a bilingual Bengali-English question-answering dataset that demonstrated considerable thinking differences between the two languages. Singh *et al.* [14] further multilingual research by developing IndicGenBench, a generative benchmark encompassing 11 Indic languages, indicating that contemporary multilingual systems still face significant hurdles in addressing morphological and syntactic variances. Shukla *et al.* [15] introduced IndicMMLU-Pro, a multilingual reasoning benchmark, and found that Indic LLMs continue to lag English-centric models in comprehension tests. Expanding resources for Bengali NLP. Fahim *et al.* [16] presented BanglaTLit, a transliteration dataset with 42.7K Romanized Bengali items, in which fine-tuned encoder models reached cutting-edge performance. Haider *et al.* [17] introduced BANTH, a multi-label hate-speech dataset for transliterated Bengali that significantly increased zero-shot classification accuracy. Romim *et al.* [18] assembled BD-SHS, a dataset of 50,000

online hate-speech samples, and achieved a 91% F1-score using contextual learning approaches. Nayeem *et al.* [19] proposed MultiBanFakeDetect, a multimodal approach for detecting false news that combines textual and visual data to improve cross-domain robustness. Furthermore, Hasan *et al.* [20] developed Bangla_ABSA, the first aspect-based sentiment analysis dataset for Bengali, which covers domains such as restaurants, mobile devices, and autos, allowing for fine-grained polarity recognition.

Nahin *et al.* [21] continued the advancement in large-scale Bengali modeling by introducing TituLLMs, a family of Bengali large language models trained on 37 billion tokens that outperformed multilingual models such as LLaMA and Zephyr in reasoning and comprehension tasks. Faria *et al.* [22] compared GPT-3.5 Turbo and Gemini 1.5 Pro to transformer-based baselines for hate-speech recognition and achieved 98.5% accuracy in a few shots. Hasan *et al.* [23] explored zero-shot and few-shot prompting for sentiment analysis and discovered that in-context examples greatly improve LLMs' generative capacities. Rafid *et al.* [24] compared decoder-only and encoder-based models for multi-label document categorization and concluded that decoder architectures respond better to unstructured textual input. Likewise, Saha *et al.* [25] established a hybrid knowledge-driven model that identifies altered Bengali news using language signals and semantic embeddings, with an accuracy of more than 90%.

C. DOMAIN-SPECIFIC LLM ADAPTATIONS AND APPLICATIONS

Hossain *et al.* [26] proposed BanglaNewsClassifier, a hybrid stacking ensemble that uses SVM, Random Forest, and XGBoost models to classify 118K Bengali news articles with 94% accuracy. Chowdhury *et al.* [27] demonstrated DepGPT, a fine-tuned large language model designed to detect depressive language in social media content, with an accuracy of 97.9%, demonstrating how LLMs may efficiently capture emotional and psychological patterns in text. Kamruzzaman *et al.* [28] generated BanMANI, an annotated dataset for detecting altered Bengali social media news, underlining the significance of domain-specific data resources for misinformation identification. Faria *et al.* [29] evaluated the performance of large language models (GPT-3.5, Gemini 1.5 Pro) and transformer-based architectures (BanglaBERT, DistilBERT) on Bengali natural language inference, revealing significant variations in reasoning capacities. Their findings showed that few-shot prompting considerably improved reasoning abilities over standard fine-tuning methods. Maity and Deroy *et al.* [30] investigated prompt-tuning strategies for Bengali grammatical error explanation using GPT-4, GPT-3.5, and LLaMA-2 and found significant increases in both accuracy and interpretability, indicating a potential direction for educational NLP applications in Bengali.

Overall, this research show Bengali NLP's rapid progress in model construction, dataset development, and cross-domain

evaluation. Earlier works were primarily concerned with resource construction and baseline transformer models; however more current research has switched to rapid engineering, multilingual adaptability, and task-specific implementations. Despite these developments, Bengali NLP continues to confront issues like as data asymmetry, inadequate tokenization, and a lack of fine-tuning resources. Building on these advances, the current study conducts a thorough evaluation of open-source LLMs for multi-domain Bengali text categorization under zero-shot and few-shot prompting scenarios, revealing novel insights into scalable and resource-efficient modelling for low-resource languages.

III. METHODOLOGY

This research performs four types of Bengali text classification utilizing six open-source Large Language Models (LLMs). The six LLMs include LLaMA-3.2-3B (Meta AI), Mistral-V3-7B (Mistral AI), DeepSeek-R1-8B (DeepSeek), Phi-4-14B (Microsoft), Gemma-2-27B (Google DeepMind), and Qwen-2.5-72B (Alibaba Cloud). The four Bengali text classification tasks were sentiment analysis, emotion recognition, hate speech detection, and fake news detection.

The reason these six LLMs were evaluated is that they have diverse training methods and were designed in different ways. They also include both smaller models that were multilingual and larger models trained to follow instructions and optimized using reinforcement learning. This variation in models facilitates equal comparison among the models based on their performance in understanding the Bengali language.

The methodology of this research is presented in the diagram, Fig.2. Initially, the Bengali text classification datasets were prepared from multiple domains, which include sentiment analysis, emotion recognition, hate speech detection, and fake news detection. The models separately undergo a zero-shot and a few-shot experimental setup. It mainly helps understand the models' in-context learning ability. The models were provided with the input sample and task instructions in case of the *zero-shot* setup. And for the *few-shot* setup, before the test input, five labeled Bengali text samples were added to the prompt, enabling the model to determine contextual relationships. The purpose of this setup is to assess whether the sample-based context enhances the performance for low-resource languages.

A. DATASETS AND TASKS

This research aims to evaluate Large Language Models based on their performance in Bengali text classification. To properly assess the models, four types of text classification were divided into two categories: opinion-based judgment (sentiment analysis and emotion recognition) and fact-based verification judgment (hate speech detection and fake news detection). The detailed information on the datasets is displayed in Table 1.

TABLE 1: Summary of datasets used in this study.

Task	Dataset	Classes	Test Size
Sentiment Analysis	SentiNoB [31]	3	1,586
Emotion Recognition	BEemoC [32]	6	625
Hate Speech Detection	Bengali HS [33]	2	1,000
Fake News Detection	BanFakeNews [34]	2	Balanced

1) Sentiment Analysis

The SentiNoB [31] dataset was used for sentiment analysis. This dataset consists of three categories collected from social media and annotated into positive, negative, and neutral. The categories contain 1586 samples, where positive was 654, negative was 571, and neutral was 361.

2) Emotion Recognition

The BEemoC dataset [32] was used for emotion recognition analysis. The dataset was annotated into six categories: joy, anger, disgust, sadness, surprise, and fear. As usual, this dataset was gathered from social networking sites. The test set comprises 625 samples.

3) Hate Speech Detection

The Bengali Hate Speech Dataset [33] was used to evaluate the hate speech detection capability of models. The binary classification framework was implemented for uniformity. A balanced dataset proportion can be viewed in the test set, which consists of 500 samples of hateful and 500 samples of non-hateful samples.

4) Fake News Detection

Fake news detection was assessed using the BanFakeNews dataset [34]. The dataset consists of almost 20,000 real news articles and 254 fake news articles. Since it is quite imbalanced, downsampling was used on real news samples to make the test set more balanced. We avoided oversampling methods such as text duplication or synthetic data generation to prevent introducing artificial patterns into the evaluation, and downsampling offered a balanced test set that allowed fair and consistent comparison across all LLMs.

B. MODELS

In this research experiment was conducted on six Large Language Models (LLMs). The models are chosen based on their variations in design, size, and training method. Additionally, models also vary based on their parameters, starting from smaller models that consist of 3B-parameter systems, to bigger multilingual models with 72B-parameter systems. The models also vary based on their framework; some models include dense Transformers, and some models were based on reinforcement learning. All these model differences together present a diversified view of models' performance in Bengali text classification.

The overall specifications of the evaluated models are summarized in Table 2.

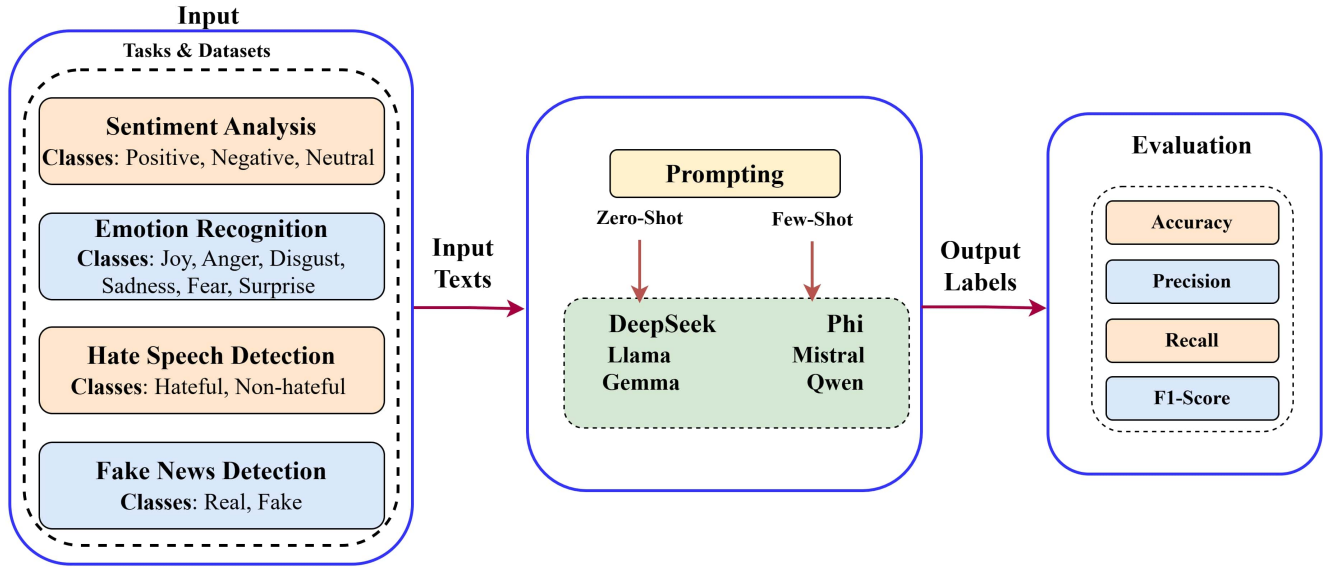


FIGURE 2: Overall methodology illustrating the evaluation framework for six open-source LLMs across four Bengali text classification tasks under zero-shot and few-shot prompting

TABLE 2: Overview of Evaluated Large Language Models.

Model	Parameters	Architecture	Developer
LLaMA-3.2-3B	3B	Transformer	Meta AI
Mistral-V3-7B	7B	Transformer	Mistral AI
DeepSeek-R1-8B	8B	Transformer + RL	DeepSeek
Phi-4-14B	14B	Transformer	Microsoft
Gemma-2-27B	27B	Transformer	Google DeepMind
Qwen-2.5-72B	72B	Transformer	Alibaba Cloud

The six selected models for assessment are based on the Transformer decoder architecture [35], where the self-attention mechanism is the central unit of operation.

The computation of the self-attention mechanism for the input sequence is as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Here, Q denotes query, K denotes key, and V denotes value metrics, and the key-vector dimension is d_k . This is improved by a multi-head attention mechanism, through the computation of attention in diverse representational layers:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (2)$$

Here, the single head computes as:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V). \quad (3)$$

Table 3 outlines the characteristics of six LLMs, including the architectural specification, the depth of layers, training parameters, and hidden dimensions.

1) LLaMA-3.2-3B

The LLaMA (Large Language Model Meta AI) series was developed by Meta [36]. It provides open and effective models for the use of multiple languages. In this series, the smallest version is the 3.2B version, and it is the most compact model with competence suitable for multiple languages, and for environments with limited resources. LLaMA-3.2-3B consists of a conventional transformer decoder structure with 28 layers. These individual layers include multi-head self-attention, which is accompanied by a feed-forward network.

A variation of multi-head attention named Grouped-Query Attention (GQA) [42] is implemented in this model. Grouped-Query Attention (GQA) preserves the multiple query heads, while reducing the number of key-value heads. The mathematical expression is as follows:

$$\text{GQA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (4)$$

Here, the queries are organized into h heads, and g set refers to a bundle of keys and values, where $g < h$. For LLaMA-3.2-3B, these queries consist of 28 query heads and 4 key-value bundles. This design decreases the memory requirements during the predictions, while ensuring the accuracy of the model, just like a full multi-head attention.

The SwiGLU activation function used in each layer of the feed-forward network is as follows:

$$\text{FFN}(x) = (\sigma(xW_1) \odot xW_2)W_3 \quad (5)$$

Here, $\sigma(x) = x \cdot \text{sigmoid}(x)$, where σ refers to the SiLU (also known as Swish) activation function, and $\odot$ refers to the

TABLE 3: Detailed Architectural Specifications of Evaluated Large Language Models.

Model	Layers (L)	Attention Heads (h)	Hidden Dim (d_h)	Context Length	Vocab Size	Training Tokens
LLaMA-3.2-3B	28	28	3,072	8,192	128,256	15T
Mistral-V3-7B	32	32	4,096	32,768	32,000	7T
DeepSeek-R1-8B	30	32	4,096	4,096	100,000	2T
Phi-4-14B	40	32	5,120	16,384	32,064	1.5T
Gemma-2-27B	46	64	8,192	8,192	256,000	13T
Qwen-2.5-72B	80	64	8,192	32,768	151,936	18T

element-wise multiplication. The dimension of intermediate feed-forward is $d_{ff} = 8192$, which is almost $2.67 \times d_h$.

LLaMA-3.2-3B was pre-trained on approximately 15 trillion text units from a huge multilingual text compilation, which covers more than 100 languages. The model utilizes the Byte-Pair-Encoding (BPE) tokenizer, which consists of 128,256 tokens maintaining the proper handling of complex scripts like Bengali. For positional encoding, the model uses Rotary Position Embeddings (RoPE), as follows:

$$\text{RoPE}(x_m, m) = \begin{bmatrix} \cos(m\theta) & -\sin(m\theta) \\ \sin(m\theta) & \cos(m\theta) \end{bmatrix} \begin{bmatrix} x_m^{(1)} \\ x_m^{(2)} \end{bmatrix} \quad (6)$$

Here, m denotes the position index and θ denotes the frequency parameter. This encoding improves the model's capability to scale to longer sequences.

The overall parameter count is approximately as follows:

$$N_{\text{params}} = L(4d_h^2 + 2d_h d_{ff}) + V d_h \approx 3.2 \times 10^9 \quad (7)$$

Here, L refers to 28 layers, $d_h = 3072$ refers to 3072 hidden dimensions, $d_{ff} = 8192$ refers to 8192 feed-forward dimensions, and V refers to 128,256 vocabulary size. Even though LLaMA-3.2-3B is on the smaller side, it still displays remarkable performance, making it ideal for zero-shot and few-shot multilingual evaluation.

2) Mistral-V3-7B

Mistral is a lightweight and effective large language model created by MistralAI; it contains 7 billion parameters [37]. The architecture is designed in such a way that it balances the prediction speed, reasoning capability, and computational efficiency. The original Mistral structure was improved with multiple enhancements to generate the third-generation model, Mistral-V3-7B. The most notable improvement was the Sliding Window Attention (SWA) mechanism, which helps with the effective processing of lengthy sequences.

The Sliding Window Attention (SWA) mechanism restricts each token's attention computation to a specific window of size W :

$$\text{SWA}(Q, K, V)_i = \text{softmax} \left(\frac{Q_i K_{[i-W:i+W]}^T}{\sqrt{d_k}} \right) V_{[i-W:i+W]} \quad (8)$$

Here, the position index is i , and $[i - W : i + W]$ refers to the adjacent tokens that are near W . Mistral utilized $W(4096)$, maintains local coherence while enabling effective long-context modeling. This helps decrease the processing complexity from $O(n^2)$ to $O(n \cdot W)$, the n refers to the length of the sequence.

The structure is made up of 32 Transformer layers, a single layer contains 32 attention heads and 4096 hidden dimensions. The SwiGLU activation is implemented in the feed-forward network, where the intermediate dimension is as follows:

$$d_{ff} = 14,336 \approx 3.5 \times d_h \quad (9)$$

Rotary Positional Embeddings (RoPE) are configured for the coherence with the sliding window pattern. The tokenizer consists of 32,000 tokens, which are built on parts of sentences. Initially augmented for the European languages, with slight inclusion of South Asian languages.

Around 7 trillion tokens are used to train the model; these tokens are gathered from various sources like: academic text, codes, and other texts. The instruction tuning and reinforcement learning from human feedback (RLHF) are employed after the model undergoes pretraining. This helps improve the model's performance based on its ability to follow instructions. The reward function of RLHF are as follows:

$$r(x, y) = \log \sigma(r_\theta(x, y_w) - r_\theta(x, y_l)) \quad (10)$$

Here, r_θ denotes the trained reward model, y_w is the desired (winner) outcome, and y_l is the less-desired (loser) outcome. Training the model in this manner enables the model to create more appropriate, accurate, and in-context responses.

3) DeepSeek-R1-8B

DeepSeekAI created an 8-billion-parameter model named DeepSeek-R1 to improve the inference and multilingual proficiency [38]. The DeepSeek-R1 model merges the large-scale pretraining with reinforcement learning from human feedback (RLHF), to enhance the inference capability, cross-language scalability.

The design is a conventional Transformer decoder design, which includes 32 attention heads, 30 layers, and 4096 hidden dimension. DeepSeek-R1 stands out from other traditional models due to having a two-segment training roadmap. In Segment 1, the model goes through conventional

next-token prediction, with the intention of causal language modeling, as the following equation:

$$\mathcal{L}_{\text{LM}} = - \sum_{t=1}^T \log P(x_t | x_{<t}; \theta) \quad (11)$$

Here, x_t denotes token at step t , $x_{<t}$ refers to the former tokens, and θ refers the model attributes.

In Segment 2, the model is streamlined by reinforcement learning (RL) via the use of an inference-oriented rewards system. The main intention for reinforcing learning (RL) is as follows:

$$\mathcal{L}_{\text{RL}} = -\mathbb{E}_{y \sim \pi_{\theta}(y|x)} [r(x, y) - \beta \text{KL}(\pi_{\theta}(y|x) || \pi_{\text{ref}}(y|x))] \quad (12)$$

Here, the reward designated to the output y is $r(x, y)$ for the input x , π_{θ} refers to the policy that is undergoing training, π_{ref} denotes the reference policy (the pre-RL model), and β ensures the model stability by managing the divergence penalty.

The reward model $r(x, y)$ consists of various elements to enhance the factual correctness, logical coherence, and the entirety of the reasoning. The reward model $r(x, y)$ is as follows:

$$r(x, y) = \alpha_1 r_{\text{correct}}(x, y) + \alpha_2 r_{\text{coherent}}(x, y) + \alpha_3 r_{\text{complete}}(x, y) \quad (13)$$

Here, r_{correct} refers to factual accuracy, r_{coherent} assesses the logical uniformity, and r_{complete} maintains that the reasoning steps are fully completed. The tuning of the coefficients α_1 , α_2 , and α_3 are done during the training to stabilize the objectives.

The DeepSeek-R1-8B was previously trained on approximately 2 trillion tokens utilizing a single tokenizer consisting of 100,000 tokens. This pretraining enables the model to effectively manage multiple languages. Rotary Positional Embedding (RoPE) is employed by the model for position encoding, along with RMSNorm:

$$\text{RMSNorm}(x) = \frac{x}{\text{RMS}(x)} \cdot \gamma, \quad \text{RMS}(x) = \sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2} \quad (14)$$

Here, γ refers to the trainable scale attribute, and the dimensionality of x is denoted by d .

4) Phi-4-14B

The Phi models are generated by Microsoft to focus on the quality of data and compilation, along with scope of data [39]. The Phi-4-14B models are a combination of size and computational efficacy. Additionally, this model presents noteworthy outcomes in structured identification and reasoning problems. The structure of Phi-4-14B consists of a conventional Transformer design, which includes 40 layers, 32 attention heads and 5120 hidden dimensions.

The principal alteration in Phi-4 is found in its data processing method rather than the architecture of the model. The Phi-4-14B model is trained with the intention of utilizing the basic causal language modeling:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^{T_i} \log P_{\theta}(x_t^{(i)} | x_{<t}^{(i)}) \quad (15)$$

Here, N denotes the serial of the training sequence and T_i refers to the length of a single sequence i .

The training data collection of Phi-4-14B consists of approximately 1.5 trillion tokens, all these tokens undergoes detailed assessment and filtering. Single record obtains a composite quality score $Q(d)$, which is calculated as follows:

$$Q(d) = w_1 q_{\text{edu}}(d) + w_2 q_{\text{fact}}(d) + w_3 q_{\text{diverse}}(d) - w_4 q_{\text{toxic}}(d) \quad (16)$$

In the equation, q_{edu} represents educational worth, q_{fact} represents logical density, q_{diverse} analyses the lexical distinctiveness, and q_{toxic} represents unfavorable or biased content. Individual records that surpass the threshold value of $Q(d) > \tau$ becomes part of the training dataset.

Furthermore, Phi-4-14B amalgamate high-quality artificial data created by bigger models and verified by human specialists. The procedure is shown as follows:

$$d_{\text{syn}} = \text{LLM}_{\text{large}}(p_{\text{template}}, k_{\text{knowledge}}) \rightarrow \text{Human}_{\text{verify}} \quad (17)$$

Here, p_{template} refers to the predefined prompt pattern, and $k_{\text{knowledge}}$ refers to compiled logical information. Human authentication maintains the logical uniformity and reliability of all the organized data.

The tokenizer used by Phi-4-14B consists of 32,064 tokens, and enables a context length of about 16,384 tokens. The proportions of the feed-forward network are as follows:

$$d_{ff} = 4 \times d_h = 20,480 \quad (18)$$

The estimated attribute count is given by the following:

$$N_{\text{params}} = L \times (4d_h^2 + 2d_h d_{ff}) + V \times d_h \approx 14 \times 10^9 \quad (19)$$

5) Gemma-2-27B

Gemma-2 is a multilingual large language model formulated by Google DeepMind. The model consist of 27 billion parameters [40]. The model was trained and fine-tuned based on a distinctive combination of data, compiled from the web, programming codes, and academic documents. The Gemma-2-27B was developed by Google to showcase a combination of an openly released model with high accuracy and effective multilingual inference.

The model design consists of Transformer layers of 46, along with attention heads of 64, and a 8192 hidden dimension. Gemma-2-27B also employs Multi-Query Attention (MQA). It is an improved version of grouped-query

attention, where the attention heads all partake in one set of keys and values. The process is as follows:

$$\text{MQA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (20)$$

Here, a single head is computed as follows:

$$\text{head}_i = \text{Attention}(Q_i W_i^Q, K, V) \quad (21)$$

The architecture decreases the memory requirement for the key-value pair caching when undergoing prediction, which enables the model to analyze the lengthy inputs with enhanced efficacy.

Rotary Positional Embeddings (RoPE) is applied by Gemma-2-27B for the purpose of position encoding while utilizing altered base frequency:

$$\theta_i = 10000^{-2i/d_h}, \quad i \in [0, d_h/2) \quad (22)$$

The feed-forward network utilizes the SwiGLU activation function with the shown proportions:

$$d_{ff} = \left\lceil \frac{8}{3} \times d_h \right\rceil \times 2 = 43,690 \quad (23)$$

The doubling factors depict the presence of a gating mechanism in SwiGLU.

The Gemma-2-27B model was trained with approximately 13 trillion tokens, that are meticulously gathered from multilingual compilations. The tokenizer includes 256,000 tokens, and it is based on SentencePiece, which enables diversified reach over Latin and non-Latin scripts. The approximate Bengali vocabulary are:

$$V_{\text{Bengali}} \approx 0.08 \times V_{\text{total}} \approx 20,480 \text{ tokens} \quad (24)$$

The broad vocabulary helps effective encoding and recognition of Bengali text, when it is differentiated from the models that mainly deal with Latin languages.

To enhance the training of multilingual, Gemma-2-27B initiates a weighted language loss function:

$$\mathcal{L}_{\text{total}} = \sum_{l \in \mathcal{L}} \alpha_l \mathcal{L}_l \quad (25)$$

Here, $\mathcal{L}_l$ denotes the deficit for each language l , and α_l denotes the temperature-managed sampling weight, which is described as follows:

$$\alpha_l = \left(\frac{n_l}{\sum_{l'} n_{l'}} \right)^{1/T} \quad (26)$$

In this equation, n_l refers to the data amount for language l , and T denotes the temperature attribute, which is usually set at 0.7. It helps improve the weight of low-resources languages.

6) Qwen-2.5-72B

The most substantial model implemented in this evaluation is Qwen-2.5, which was created as a segment of Qwen2.5 series by Alibaba Group [41]. The model is made up of 72 billion parameters and was trained on a huge multilingual compilation consisting of about 18 trillion tokens. The compilations also consist of various Indic languages, like Bengali. The Qwen-2.5 is considered to be most extensive open-resourced multilingual models that is accessible.

The structure of the model consists of 80 layers of Transformers along with 64 attention heads, and 8194 hidden dimensions. The model utilizes Grouped-Query Attention (GQA) [42] with 8 key value group, it helps enhance the runtime capacity and memory effectiveness. The setup is as follows:

$$h_{\text{query}} = 64, \quad h_{\text{kv}} = 8, \quad \text{ratio} = \frac{h_{\text{query}}}{h_{\text{kv}}} = 8 \quad (27)$$

The attention calculation for a single group can be written as:

$$\text{head}_{i,g} = \text{Attention}(Q_i W_i^Q, K_g W_g^K, V_g W_g^V) \quad (28)$$

Here, i is the query heads index and $g = \lfloor i/8 \rfloor$ chooses the related key-value group. The configuration of the feed-forward networks is as follows:

The feed-forward network dimension is given by:

$$d_{ff} = \frac{256}{3} \times d_h = 69,632 \quad (29)$$

An estimation of the total attribute are:

$$\begin{aligned} N_{\text{params}} &= L \times (4d_h^2 + 2d_h d_{ff}) + V \times d_h \\ &= 80 \times (4 \times 8192^2 + 2 \times 8192 \times 69632) \\ &\quad + 151,936 \times 8192 \\ &\approx 72 \times 10^9 \end{aligned} \quad (30)$$

Qwen-2.5 -72B employs the byte-level BBPE tokenizer, which consists of a vocabulary of 151,936 tokens [41]. It is estimated that 12% of the tokens are assigned to the Indic language, hence Bengali sub-vocabulary gets around:

$$V_{\text{Bengali}} \approx 0.12 \times V_{\text{total}} \approx 18,232 \quad (31)$$

The training roadmap is divided into three parts, which incorporate the general pretraining, instruction fine-tuning, and domain continued pretraining:

$$\mathcal{D}_{\text{train}} = \mathcal{D}_{\text{general}} \cup \mathcal{D}_{\text{domain}} \cup \mathcal{D}_{\text{instruction}} \quad (32)$$

To enhance the depiction of low-resource languages like Bengali, a sampling strategy was also implemented:

$$p(l) = \frac{(n_l/N)^\alpha}{\sum_{l'} (n_{l'}/N)^\alpha}, \quad \alpha = 0.3, \quad (33)$$

Here, n_l refers to the token numbers for language l and N is the total amount of tokens.

C. PROMPTING STRATEGIES

To analyze the in-context learning competence of the applied Large Language Models (LLMs), two prompting techniques were implemented during the experiment.

Zero-shot prompting: For the zero-shot approach, a task instruction prompt is given to the model, for example detect the sentiment or emotion. Additionally, no more samples or context are provided with the prompt. Following this, models utilize their general knowledge initiated completely from the provided prompt to create responses for task completion.

Few-shot prompting (five-shot): For the few-shot approach, the model is not provided with any instructions; instead, it is given five examples along with the outputs of the examples. Then the model proceeds to generate responses by deducing the task criteria from the five examples. This enhances the accuracy of models in solving the tasks.

The structure of these techniques adheres to the previous papers on in-context learning [35], [43], because previously, after giving the models a few samples, the models performed better and adjusted to the new data more, specifically in the case of low-resource language situations.

Task-related prompts were generated for Bengali text classification, the four types of classifications include *Sentiment Analysis*, *Emotion Recognition*, *Hate Speech Detection*, and *Fake News Detection*. The prompt was drafted individually to properly describe the task, explain the label space, and stipulate the necessary JSON output pattern for uniform parsing and assessment.

This consistent pattern enables equal comparison among the LLMs. We differentiate the zero-shot and few-shot systems to analyze if the incorporation of contextual samples enhanced the classification performance and accuracy for Bengali text classification.

D. EXPERIMENTAL SETUP

All model evaluations were carried out using the *vLLM* library¹, which offers efficient memory management for large language models. The experiments were run on a local machine with an AMD Ryzen 5 8400F (4.20 GHz, 6-core) processor, 16 GB RAM, and an NVIDIA GeForce RTX 4060 GPU (8 GB VRAM).

Very large models such as Qwen-2.5-72B cannot be loaded directly on an 8 GB GPU. To handle this, we used a 4-bit quantized version of the model (AWQ/GPTQ) along with CPU-GPU offloading supported by *vLLM*. In this approach, most of the weights remain in system memory, and only the necessary parts are temporarily moved to the GPU during inference. This method allowed us to run the model on limited hardware without affecting the output noticeably. To confirm this, we compared a small set of results with an FP16 version of the same model on a cloud A100 GPU and found no meaningful differences in prediction labels.

All models were tested under the same decoding settings. The temperature was set to 0 to ensure deterministic outputs,

TABLE 4: Inference Configuration Parameters.

Parameter	Value
Temperature	0
Top- <i>p</i>	0.8
Repetition Penalty	1.05
Max Tokens	512
GPU	NVIDIA GeForce RTX 4060 (8 GB)
Processor	AMD Ryzen 5 8400F (4.20 GHz, 6-Core)
System RAM	16 GB
Quantization	4-bit AWQ/GPTQ
vLLM Features	PagedAttention + CPU Offload

and the top-*p* value was fixed at 0.8. A repetition penalty of 1.05 was applied to reduce repeated patterns. For every task, the maximum generated output length was set to 512 tokens. These consistent settings ensured that any variation in performance came from the models themselves and not from differences in generation parameters.

All experiments were carried out on a 64-bit Windows 11 system with CUDA and PyTorch properly configured. By combining *vLLM*'s memory-efficient scheduling with 4-bit quantization, we were able to evaluate all models, including those with very large parameter counts, on the same hardware in a consistent and reproducible manner.

E. EVALUATION METRICS

The models were evaluated using contemporary evaluation metrics: *Accuracy*, *Precision*, *Recall*, and *F1-score*. Precision and Recall demonstrate the relation between false positives and false negatives. The F1 score is important for an imbalanced dataset, if the dataset is imbalanced, multiple metrics are required to appropriately analyze the model's performance. The F1 score is calculated by the balanced mean of Precision and Recall. And the most vital metric, Accuracy, gives an overall rate of the model's ability to correctly identify the labels based on overall predictions [44].

Moreover, a *confusion matrix* was created for each model to gather True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN), which provides an overall view of the model's errors and biases.

The formal definition of the four-evaluation metrics is as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (34)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (35)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (36)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (37)$$

Here,

¹<https://docs.vllm.ai>

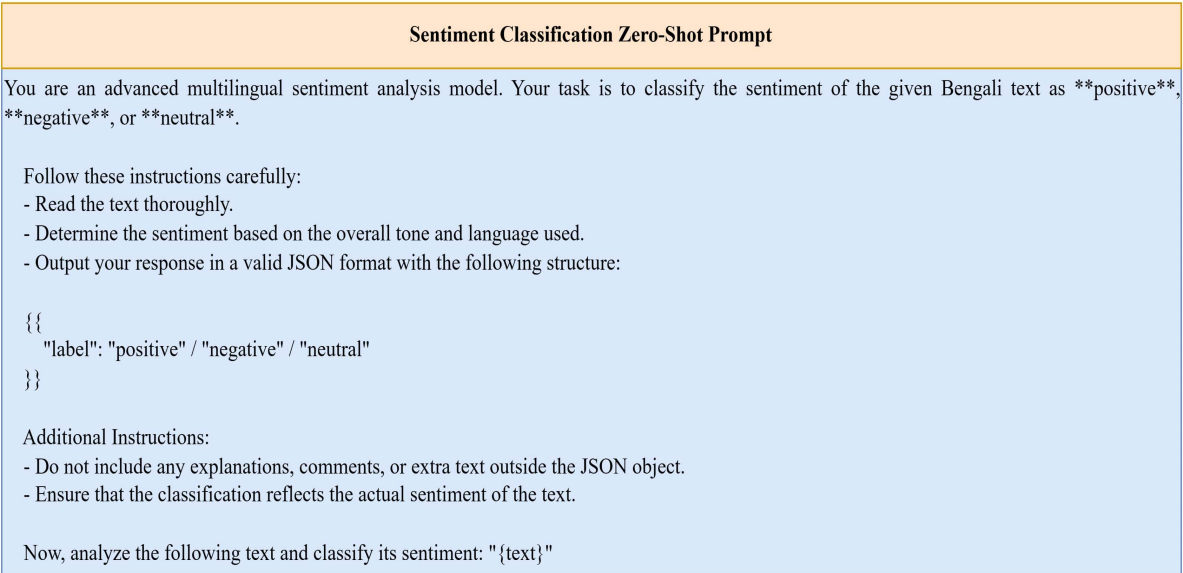


FIGURE 3: Prompt used for zero-shot Bengali Sentiment Analysis.

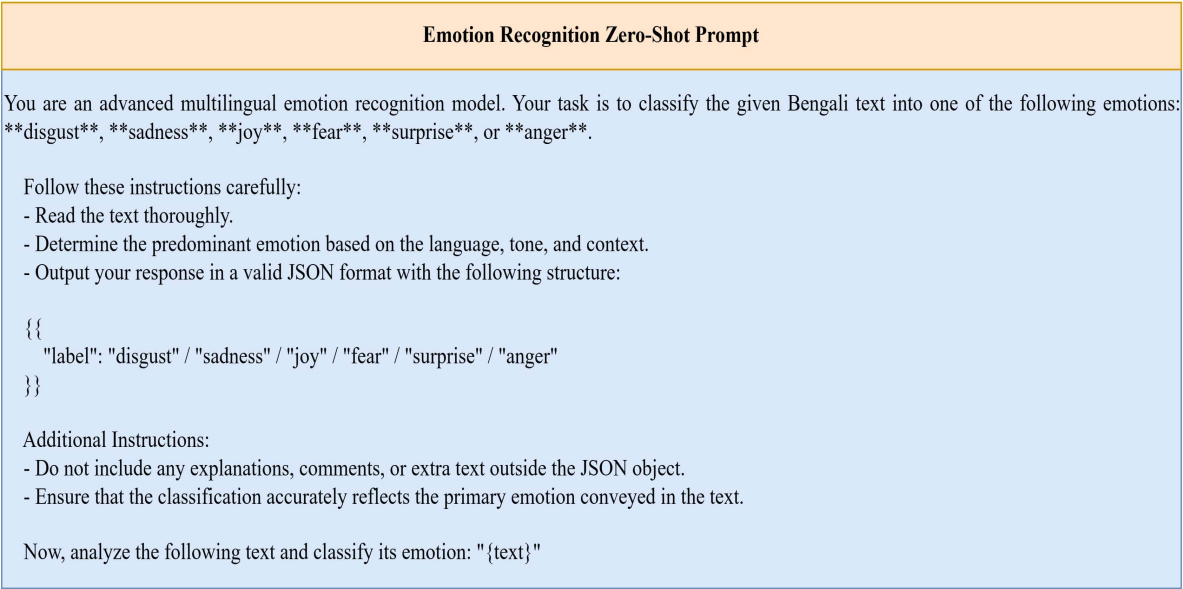


FIGURE 4: Prompt used for zero-shot Bengali Emotion Recognition.

- Accuracy calculates the overall performance of the model by taking the ratio of accurate predictions to the total number of predictions.
- Precision gives an overall view of the number of accurate positive predictions among the total positive predictions.
- Recall gives an overall view of the number of actual positive predictions that were correctly predicted.
- F1-score provides a uniform calculation between Precision and Recall, specifically for the situation when the dataset is imbalanced.

IV. RESULTS AND ANALYSIS

This section presents the performance of various LLMs on two prompting settings across four classification tasks. The associated results for each prompting strategy are described as follows.

A. ZERO-SHOT PERFORMANCE

Table 5 presents the zero-shot performance of all evaluated large language models (LLMs) across the four Bengali text classification tasks. For the sentiment Analysis task, Gemma-2-27B reported the overall best results on accuracy

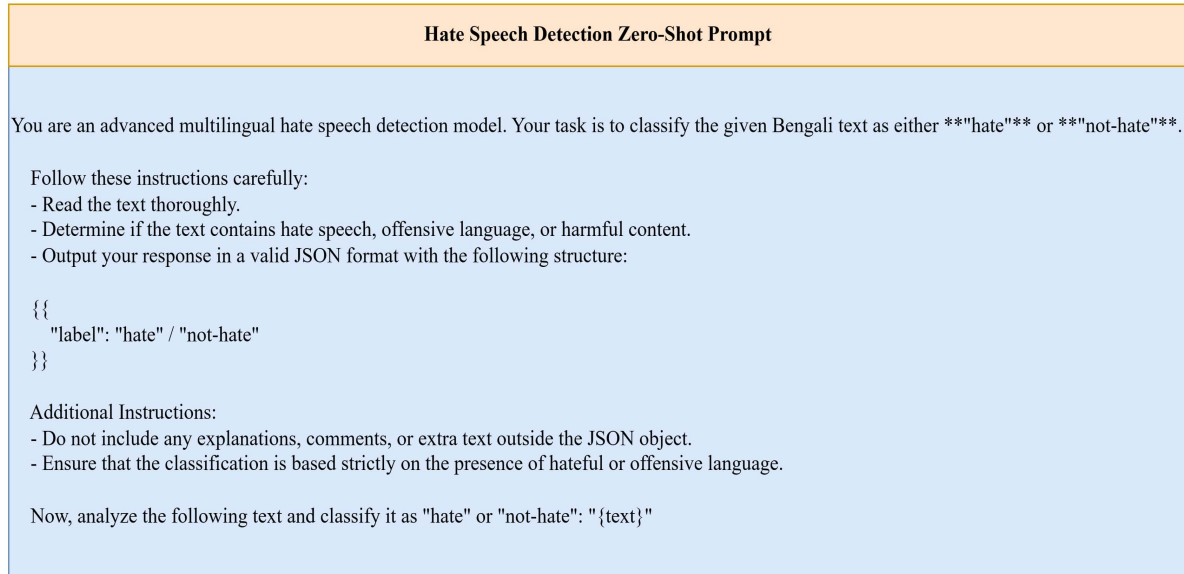


FIGURE 5: Prompt used for zero-shot Bengali Hate Speech Detection.

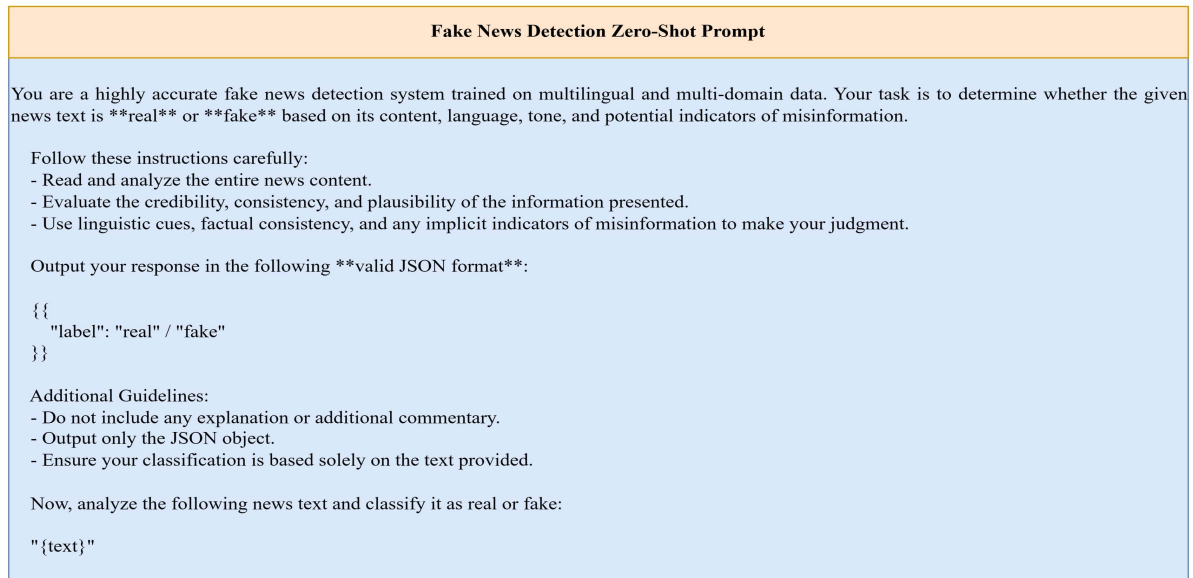


FIGURE 6: Prompt used for zero-shot Bengali Fake News Detection.

(62.04%), recall (62.04%), and F1-Score (61.54%) a word-based approach that generalizes very well even without in-context usage of examples. In comparison, Qwen-2.5-72B achieved the best precision with 66.12%, meaning it was better at suppressing false positives and generating confident predictions.

For the Emotion Recognition benchmark, we compare a combination of Qwen-2.5-72B and Gemma-2-27B exhibited comparable results consistently, with an accuracy of 61.76%. Their F1-scores of 59.44% and 58.40%, respectively, indicate that such large multilingual models have a reasonable ability to recognise emotion expressions in Bengali text without

task-specific fine-tuning.

Among the Hate Speech Detection systems, Qwen-2.5-72B achieved the best results in terms of overall performance, with an accuracy of 79.90% and an F1-score of 79.88%, demonstrating that this model is very robust in detecting offensive or harmful material. We achieved the best accuracy of 84.33% in Fake News Detection on Gemma-2-27B, indicating it is competent to separate fact from fake.

In general, the results suggest that the larger models, especially Qwen-2.5-72B and Gemma-2-27B perform better than the smaller models in all tasks. The success of

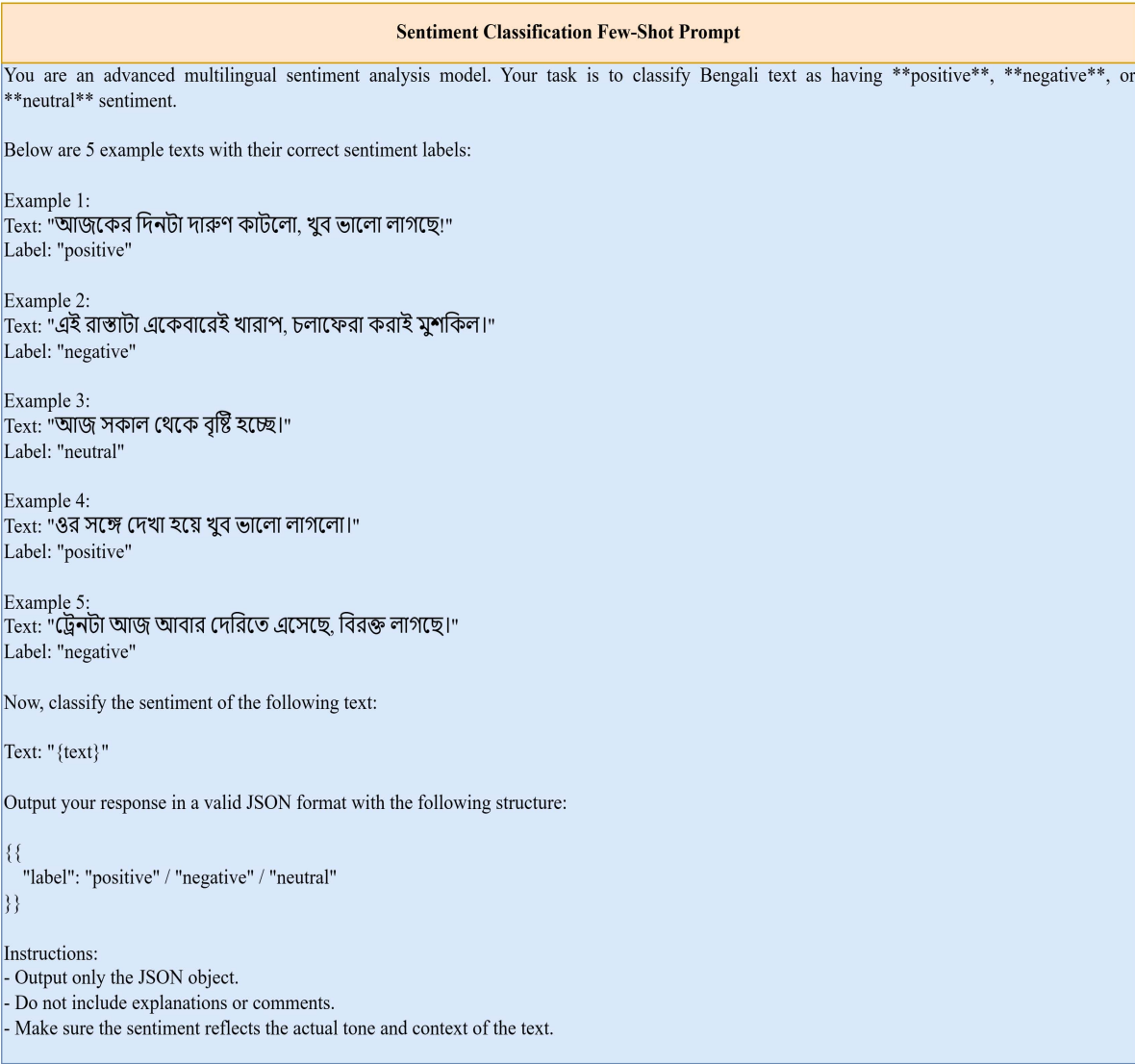


FIGURE 7: Prompt used for few-shot Bengali Sentiment Analysis.

these models is due to their larger number of parameters, multilingual training, and wider token coverage, contributing to a more comprehensive understanding of Bengali syntactic and semantic context. Smaller models such as LLaMA-3.2-3B, Mistral-V3-7B, and DeepSeek-R1-8B were less competitive. This could be partially because they have a smaller capacity and receive less Bengali exposure in pretraining, so that their generalisation is weaker for low-resource language tasks.

B. FEW-SHOT PERFORMANCE

Table 6 presents the performance of the evaluated large language models (LLMs) in the few-shot setting, where each model was provided with five labelled examples per task. Overall, Qwen-2.5-72B performs the best in the Emotion Recognition and Hate Speech Detection tasks, with an F1-score of 65.86% and 78.25%, respectively.

Gemma-2-27B achieves outstanding results in the Sentiment Analysis and Fake News Detection tasks with an F1-score of 63.00% and 84.17%. These results suggest that Qwen-2.5-72B and Gemma-2-27B are consistent well across different domains and utilise few-shot context effectively.

A significant gain is achieved by this transfer on all models with respect to the zero-shot performance. This demonstrates that modest supervision in context provides models with a better understanding of the task and a means for capturing language idiosyncrasies specific to Bengali, which might be underrepresented in large-scale pre-training corpora.

Both Mistral-V3-7B and DeepSeek-R1-8B show relatively weak performance compared with the other smaller models. For example, their F1-scores in Sentiment Analysis are 49.71% and 51.71%, while in Emotion Recognition they reach only 46.41% and 43.41%. This inferior performance can be explained partly because of the smaller parameter

Emotion Recognition Few-Shot Prompt								
You are an advanced multilingual emotion recognition model. Your task is to classify Bengali text into one of the following emotions: **disgust**, **sadness**, **joy**, **fear**, **surprise**, or **anger**. Below are 5 example texts with their correct emotion labels: Example 1: Text: "ওঁর এমন ব্যবহারে আমি ভীষণ ঘৃণা বোধ করেছি।" Label: "disgust" Example 2: Text: "আজ আমার বাবার মৃত্যুবার্ষিকী, খুব কষ্ট পাচ্ছি।" Label: "sadness" Example 3: Text: "আমি পরীক্ষায় প্রথম হয়েছি! খুব খুশি লাগছে।" Label: "joy" Example 4: Text: "রাতে হঠাৎ দরজার আওয়াজ পেয়ে খুব ভয় লাগলো।" Label: "fear" Example 5: Text: "ও এমন কথা বলবে ভাবতেই পারিনি!" Label: "surprise" Now, classify the following text into one of the predefined emotions. Text: "{text}" Output your response in a valid JSON format with the following structure: <pre>{{ "label": "disgust" / "sadness" / "joy" / "fear" / "surprise" / "anger" }}</pre> Instructions: - Output only the JSON object. - Do not include explanations or comments. - Ensure the label reflects the main emotion conveyed in the text.								

FIGURE 8: Prompt used for few-shot Bengali Emotion Recognition.

TABLE 5: Zero-shot evaluation results of Large Language Models (LLMs) on various Bengali text classification tasks.

Model	Sentiment Analysis				Emotion Recognition			
	Acc.	P.	R.	F1	Acc.	P.	R.	F1
LLaMA-3.2-3B	52.21	54.62	52.21	51.83	46.24	49.08	46.24	44.28
Mistral-V3-7B	51.32	54.49	51.32	52.25	39.86	39.40	36.96	33.89
DeepSeek-R1-8B	50.82	55.15	50.82	52.55	45.92	47.48	45.92	42.14
Phi-4-14B	59.84	61.88	59.84	60.28	59.76	59.73	59.76	56.96
Gemma-2-27B	62.04	65.12	62.04	61.54	61.76	68.47	61.76	58.40
Qwen-2.5-72B	58.07	66.12	58.07	59.32	61.76	67.67	61.76	59.44
Model	Hate Speech Detection				Fake News Detection			
	Acc.	P.	R.	F1	Acc.	P.	R.	F1
LLaMA-3.2-3B	57.10	59.49	57.10	54.22	77.76	77.95	77.76	77.72
Mistral-V3-7B	64.70	65.58	64.70	64.19	61.11	76.02	61.11	55.45
DeepSeek-R1-8B	59.30	63.97	59.30	56.40	74.30	77.36	74.30	73.91
Phi-4-14B	72.50	73.52	72.50	72.77	77.36	79.64	77.36	76.91
Gemma-2-27B	76.50	78.33	76.50	76.11	84.45	85.57	84.45	84.33
Qwen-2.5-72B	79.90	80.05	79.90	79.88	83.27	86.26	83.27	82.91

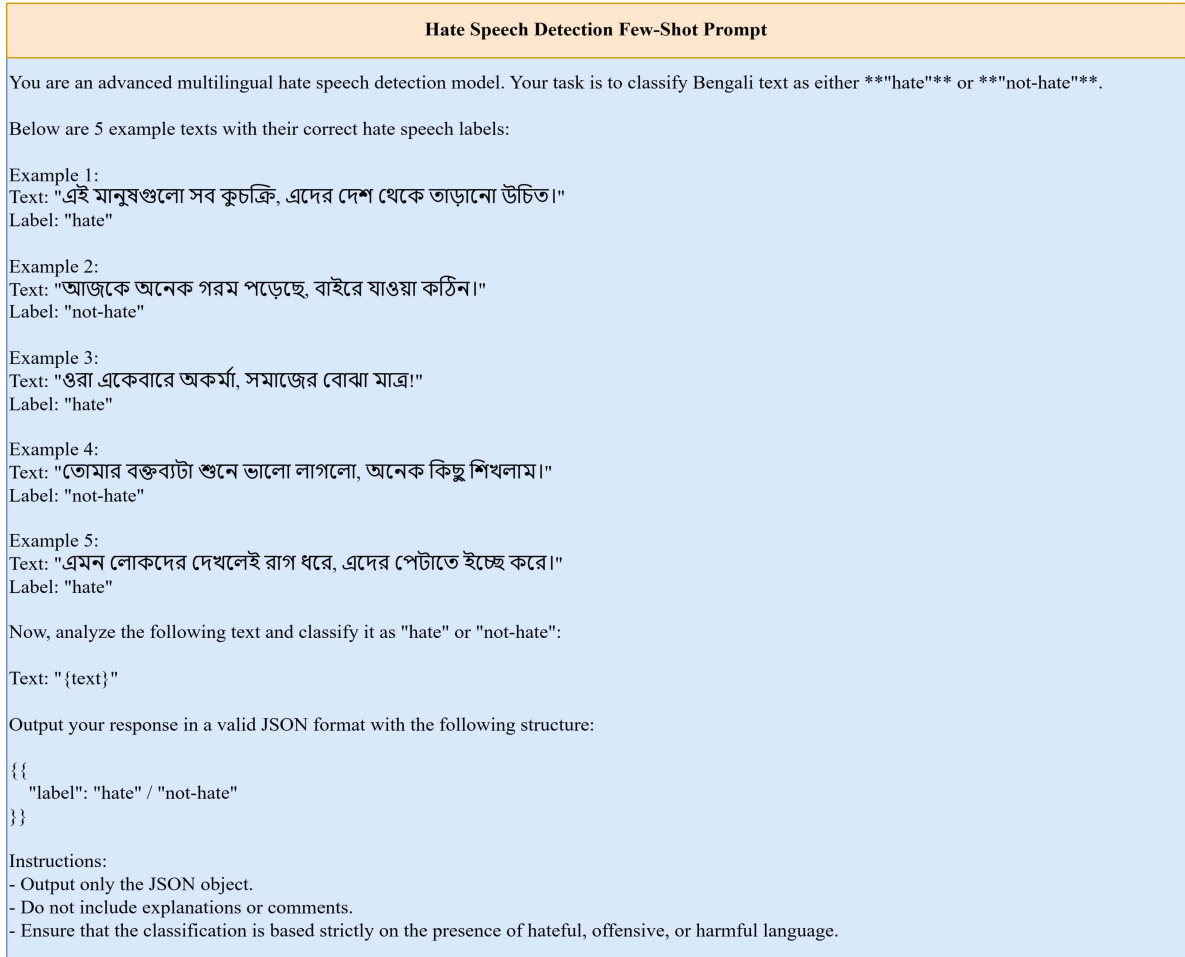


FIGURE 9: Prompt used for few-shot Bengali Hate Speech Detection.

size and less multilingual training, as well as lesser exposure to Bengali in the pre-training. A larger multilingual model, Qwen-2.5-72B and Gemma-2-27B exhibit more robust generalisation ability, verifying the importance of scale and diverse multilingual data in few-shot learning for low-resource languages.

C. CLASS-WISE PERFORMANCE ANALYSIS

This section discusses the LLMs' performance across different classes of the four classification tasks.

1) LLMs Performance Across Sentiment Categories

Fig.11 presents the F1-scores of each sentiment class across all large language models (LLMs) for both zero-shot and few-shot prompting settings. In the zero-shot condition, all models achieve higher F1 scores on the negative and positive sentiment classes when comparing with Gemma-2-27B model and Qwen-2.5-72B with the most optimal performance. However, the neutral class remains inferior in F1-score across all models, consistently showing that separating neutral sentiment is still difficult in the

absence of context examples.

The trend is similar in the few-shot scenario, with substantial improvements for the neutral class observed across several models. For example, Gemma-2-27B shows an increase of approximately five percentage points in the neutral category when moving from zero-shot to few-shot prompting. This upside reveals that few-shot prompting enables models to better understand the nuanced linguistic embedding of and context cues in the input text. Because of being exposed to these labelled examples, the models capture ambiguous or mixed sentiment expression much better, and achieve a more balanced performance across sentiment categories.

2) LLMs Performance Across Emotion Categories

Fig.12 shows the F1 scores across emotion categories. As we can see in the zero-shot setting, the performance of LLMs is highly heterogeneous on emotion types, from Joy and Fear classes with higher F1 scores to Disgust and Anger categories being more difficult to classify accurately. Qwen-2.5-72B and Gemma-2-27B again show strong

Fake News Detection Few-Shot Prompt
You are a highly accurate fake news detection system trained on multilingual and multi-domain data. Your task is to determine whether the given news text is **real** or **fake** based on its content, language, tone, and potential indicators of misinformation. Below are 5 example news texts along with their correct classifications: Example 1: Text: "সরকার ঘোষণা করেছে আগামীকাল থেকে দেশের সব শিক্ষাপ্রতিষ্ঠান পুনরায় খোলা হবে।" Label: "real" Example 2: Text: "একটি গবেষণায় দেখা গেছে যে প্রতিদিন লবঙ্গ খেলে করোনা সম্পূর্ণরূপে নিরাময় হয়।" Label: "fake" Example 3: Text: "চট্টগ্রামে ভূমিকম্পে ক্ষতিগ্রস্ত হয়েছে বহু বাড়িঘর, উদ্ধার কাজ চলছে।" Label: "real" Example 4: Text: "চাঁদের বুকে বিশালাকার প্রাণী দেখা গেছে, নাসা নিশ্চিত করেছে।" Label: "fake" Example 5: Text: "বাংলাদেশ ক্রিকেট দল আগামী মাসে ভারত সফরে যাচ্ছে বলে জানিয়েছে বিসিবি।" Label: "real" Now, analyze the following news text and classify it as **real** or **fake**: Text: "{text}" Respond in the following **valid JSON format**: <pre>{{ "label": "real" / "fake" }}</pre> Instructions: - Do not include any explanation or commentary. - Output only the JSON object. - Ensure your classification is based solely on the content and plausibility of the news text.

FIGURE 10: Prompt used for few-shot Bengali Fake News Detection.

TABLE 6: Few-shot evaluation results of Large Language Models (LLMs) on various Bengali text classification tasks.

Model	Sentiment Analysis				Emotion Recognition			
	Acc.	P.	R.	F1	Acc.	P.	R.	F1
LLaMA-3.2-3B	57.12	57.50	57.12	57.19	46.72	63.35	46.72	47.12
Mistral-V3-7B	47.92	58.03	47.92	49.71	46.24	57.79	46.24	46.41
DeepSeek-R1-8B	50.63	54.20	50.63	51.71	46.56	45.69	46.56	43.41
Phi-4-14B	58.32	64.24	58.32	58.88	56.00	64.95	56.00	54.56
Gemma-2-27B	62.55	67.93	62.55	63.00	61.12	71.66	61.12	60.27
Qwen-2.5-72B	60.28	66.86	60.28	61.30	65.76	70.79	65.76	65.86
Model	Hate Speech Detection				Fake News Detection			
	Acc.	P.	R.	F1	Acc.	P.	R.	F1
LLaMA-3.2-3B	64.40	67.28	64.40	62.85	69.29	80.05	69.29	66.27
Mistral-V3-7B	63.10	63.14	63.10	63.07	63.58	74.84	63.58	58.93
DeepSeek-R1-8B	62.30	63.46	62.30	61.47	67.32	71.69	67.32	65.59
Phi-4-14B	71.00	72.33	71.00	70.56	81.69	83.25	81.69	81.48
Gemma-2-27B	74.30	78.15	74.30	73.39	84.45	87.03	84.45	84.17
Qwen-2.5-72B	78.40	79.18	78.40	78.25	84.06	87.50	84.06	83.68

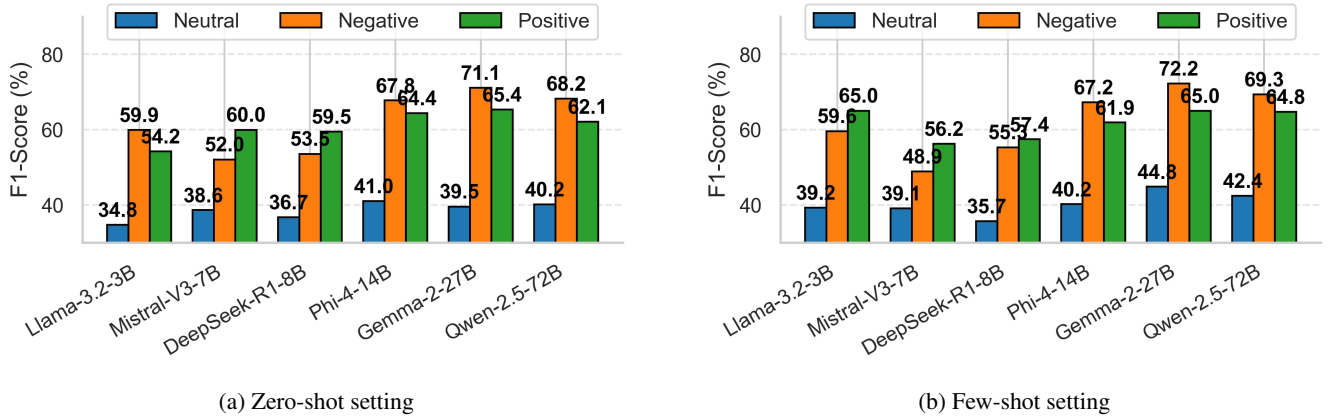


FIGURE 11: Class-wise sentiment classification performance for two prompt settings across LLMs.

performance, except in negative emotions such as Anger and Disgust. While few-shot prompting with Qwen-2.5-72B improves performance by 15% in the Disgust class compared to zero-shot prompting, it results in a 5% decrease in the Anger class. But all other emotion classes (Joy, Fear, Sadness, Surprise) show significant improvement of more than 5% than their zero-shot equivalence. We also notice that the Llama-3.2-3B, Mistral-V3-7B, and DeepSeek-R1-8B perform poorly across most of the categories, especially in Anger, Disgust, and Surprise, with both prompting approaches.

3) LLMs Performance Across Hate and Not-Hate Categories

Fig.13 presents the class-wise F1-scores for the Hate Speech Detection task under both zero-shot and few-shot prompting settings. In the zero-shot setting, almost all the models, including LLaMA-3.2-3B, struggle to identify the hate class accurately. Our analysis reveals the incapacity of such models (e.g., Qwen-2.5-72B, Mistral-V3-7B, and DeepSeek-R1-8B) to correctly identify the hate class as models obtain an F1-score under 50% in most cases. Still, their accuracy on the not-hate class is disproportionately high, suggesting that the model strongly prefers non-hateful (or safer) predictions in the absence of labelled examples. This disparity indicates that smaller-sized or less multilingual models do not generalise to socially sensitive language situations in Bengali very well.

While only a small drop can be seen in the not-hate class, the few-shot setting leads to a clear improvement in the hate class across all models. For instance, Qwen-2.5-72B achieves an F1-score of 80.6% for the hate class and 79.2% for the not-hate class. Similarly, Gemma-2-27B records 79.2% for the hate class and 73.1% for the not-hate class. These results show that few-shot prompting helps the models detect both explicit and implicit hate expressions while maintaining balanced classification. For the larger multilingual models, such as Qwen-2.5-72B and Gemma-2-27B, the difference between the zero-shot and few-shot performance is relatively small, suggesting that these models already have a strong

contextual understanding even before additional examples are provided.

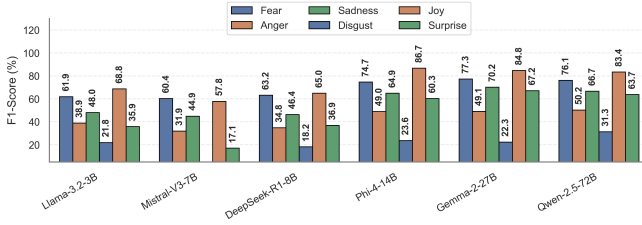
4) LLMs Performance Across Fake and Real News

Fig.14 presents the class-wise F1-scores for the Fake News Detection task under both zero-shot and few-shot prompting conditions. Under zero-shot setting, most models obtain the F1-score over 70% in both real and fake classes, which demonstrates they can generalise well without being access to any example with labels. Nonetheless, Mistral-V3-7B is particularly weak at detecting fake news, with an F1-score of only 38.9% for the fake class. The bigger guys, Gemma-2-27B and Qwen-2.5-72B to have the best performance in both classes, with more than 80% for each. This result demonstrates their superior performance in discriminating between real and fake content using the Bengali language.

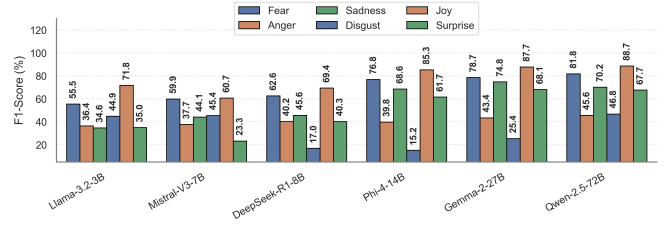
Interestingly, the few-shot setting doesn't give an obvious improvement for this task. Smaller by sizes are models, such as LLaMA-3.2-3B and DeepSeek-R1-8B obtain slight improvements, while the larger models have minor performance decrements. For instance, Gemma-2-27B exhibits F1-scores of 82.1% and 83.5%, in the case of the fake and real classes consecutively, versus 82.9% and 85.7%, obtained under the zero-shot setup. Similarly, Qwen-2.5-72B shows a marginal decrease from 80.5% to 81.2% for the fake class and from 85.4% to 86.2% for the real class. This trend hints that, differently from other tasks, presenting few-shot instances does not largely contribute to the improvement of fake news detection. One possible explanation is that the models already have strong contextual and factual reasoning ability from pretraining, which renders further examples less helpful.

D. ERROR ANALYSIS

This section analyzes the prediction behavior of the evaluated models using confusion matrices. The objective is to understand the common misclassification patterns and assess

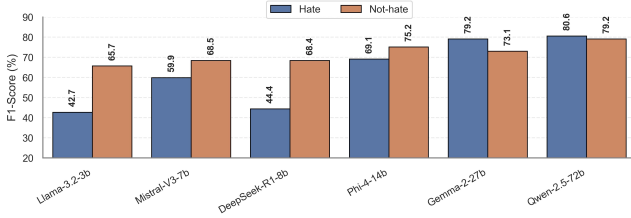


(a) Zero-shot setting

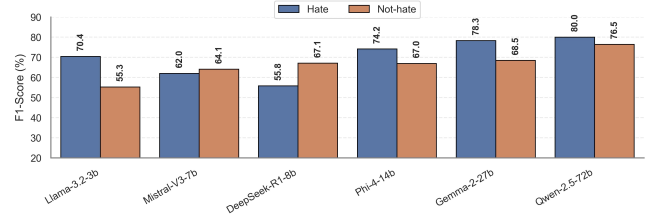


(b) Few-shot setting

FIGURE 12: Class-wise emotion recognition performance for two prompt settings across LLMs.

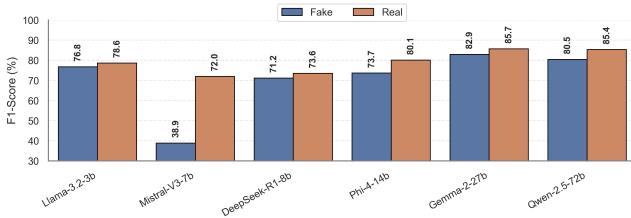


(a) Zero-shot setting

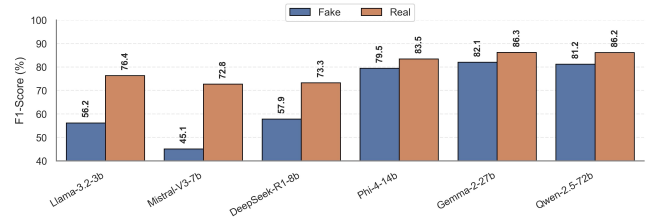


(b) Few-shot setting

FIGURE 13: Class-wise hate speech detection performance for two prompt settings across LLMs.



(a) Zero-shot setting



(b) Few-shot setting

FIGURE 14: Class-wise fake news detection performance for two prompt settings across LLMs.

how model architecture and prompting strategies affect performance across the four Bengali text classification tasks.

1) Best Performing Models

The confusion matrices in Fig. 15 show the performance of the best-performing models. In the sentiment analysis task, Gemma-2-27B is competitive in both few-shot and zero-shot settings. As shown in Fig. 15(a) and Fig. 15(b), most samples align along the diagonal, indicating accurate predictions for positive and negative sentiments. A few neutral samples are misclassified as negative, which means that sometimes the model fails to detect mild form of neutrality in the colloquial Bengali text.

For the emotion recognition task, Qwen-2.5-72B demonstrates consistent reliability across major emotion categories, as seen in Fig. 15(c) and Fig. 15(d). The model is successful in recognising joy and fear, but the identification of anger and disgust appears not to be as straightforward, possibly due to the similar linguistic cues they typically share. Virtually grounded prompting does give a small boost in performance for rare emotions such as surprise, which

suggests that there is some benefit to seeing a small number of examples from the context.

In the hate speech detection task, Fig. 15(e) and Fig. 15(f) show that Qwen-2.5-72B achieves balanced classification between hate and not-hate categories. The few-shot setting improves recall for the hate class, helping the model detect implicit or context-dependent hostility expressed in Bengali. In the fake news detection task, shown in Fig. 15(g) and Fig. 15(h), Gemma-2-27B achieves high precision for both fake and real content with minimal off-diagonal errors. This consistency demonstrates that the model effectively uses linguistic and contextual cues to separate factual information from misinformation.

Overall, the best-performing models display clear diagonal concentration with very few misclassifications, confirming their strong contextual and linguistic understanding of Bengali text. Their advantage is likely due to larger parameter counts and extensive multilingual pre-training, which improve their adaptability to low-resource languages.

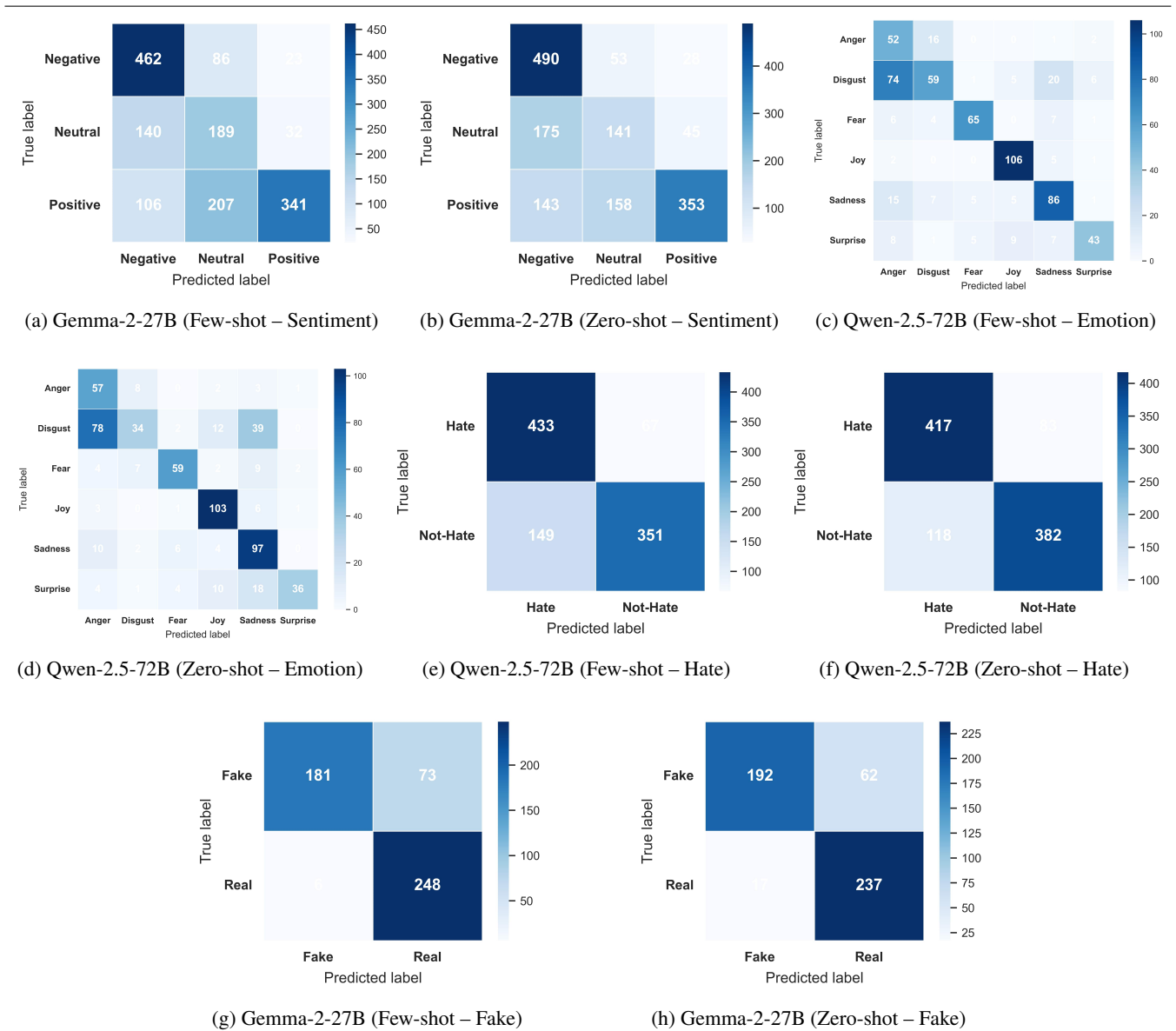


FIGURE 15: Confusion matrices of top LLMs on Bengali text classification tasks under zero- and few-shot settings.

2) Worst Performing Models

Fig. 16 presents the confusion matrices for the least effective models. In sentiment analysis, shown in Fig. 16(a) and Fig. 16(b), LLaMA-3.2-3B and Mistral-V3-7B display significant confusion between neutral and negative samples, with weak diagonal concentration. This suggests that smaller models lack sufficient semantic depth to capture subtle tonal shifts in Bengali sentiment expressions.

For emotion recognition, Fig. 16(c) and 16(d) show that Mistral-V3-7B and DeepSeek-R1-8B produce scattered distributions with frequent overlaps between sadness, disgust, and anger. The few-shot configuration yields limited improvement, suggesting that model capacity, rather than exposure to examples, limits emotional differentiation.

In the hate speech detection task (Fig. 16(e) and 16(f)),

both DeepSeek-R1-8B and LLaMA-3.2-3B misclassify many hateful statements as not-hate. This reflects a cautious prediction bias, where the models avoid labelling a text as hateful unless explicit aggression is present. Finally, for fake news detection, shown in Fig. 16(g) and Fig. 16(h), Mistral-V3-7B struggles to detect fabricated content, frequently identifying fake instances as real. These scattered prediction patterns demonstrate the limited generalization capability of smaller or less optimised models.

E. RECEIVER OPERATING CHARACTERISTIC (ROC-AUC) ANALYSIS

In this section, we evaluate the discriminative performance of the models using confusion matrix analysis, ROC curves, and AUC scores. The ROC–AUC score attempts to present a fair judgment to compare models on the positive and negative

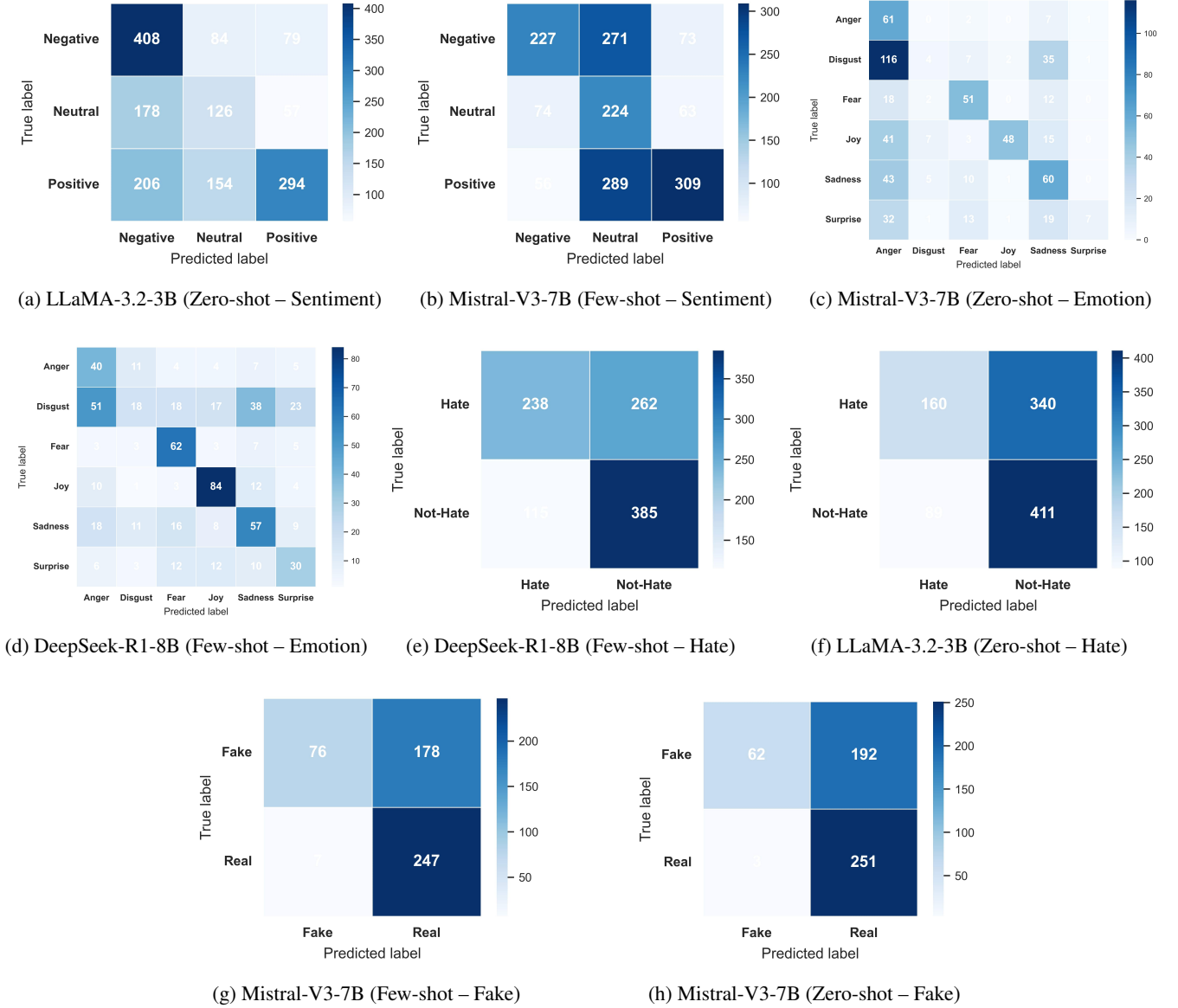


FIGURE 16: Confusion matrices of low-performing LLMs showing misclassification patterns in Bengali tasks.

classes in different prompting strategies. Fig. 17 illustrates the ROC–AUC results for all four Bengali text classification tasks under both zero-shot and few-shot conditions.

As shown in Fig. 17(a) and Fig. 17(b), the sentiment analysis task reveals moderate separability for most models, with AUC values ranging from 0.63 to 0.72. Gemma-2-27B achieves the highest AUC of 0.72 in the few-shot setting, followed closely by Qwen-2.5-72B (0.70), demonstrating more confident and stable classification boundaries. Smaller models such as LLaMA-3.2-3B and Mistral-V3-7B show weaker discrimination, indicating difficulties in capturing subtle sentiment cues in Bengali.

For the emotion recognition task (Fig. 17(c) and Fig. 17(d)), all models get a higher AUC value than that of the sentiment task. As shown in the few-shot setting, Qwen-2.5-72B and Gemma-2-27B achieve higher 0.81 and

0.80 than others, revealing their healthy discriminative power on emotional tone. While Phi-4-14B also maintains a competitive AUC at 0.77, smaller models fail to consistently capture sensitivity between groups of emotion classes.

In hate speech detection (Fig. 17(e) and Fig. 17(f)), Gemma-2-27B and Qwen-2.5-72B outperform the other ones with few-shot AUCs higher than 0.74 and 0.78, respectively. This improvement highlights the benefit of in-context learning for sensitive social classification tasks where context and tone are important factors. By contrast, LLaMA-3.2-3B gets a low AUC of only 0.57 in the zero-shot setting, indicating that LLaMA-3.2-3B is less strong in detecting offensive or hateful content.

The fake news detection task (Fig. 17(g) and Fig. 17(h)) obtains the highest AUC values in all tasks. Both Gemma-2-27B and Qwen-2.5-72B obtain 0.84 in the two

versions, prompting, indicating a good discriminative level for real and fake text. This consistent line of results further confirms that the larger multilingual LLMs are indeed more capable to identify misinformation due to their access to a wider range of facts and contexts.

V. DISCUSSION

This section provides a summary based on the observations and errors of the results, discussing some of the trends related to performance in all tasks across various LLMs under zero-shot and few-shot prompting. The tendency of the general performance is shown in Fig. 18. The easiest task among those analysed is Fake News detection, which reaches high F1-scores in all models regardless of the dataset used; Emotion Recognition remains as the toughest one. Overall, few-shot prompting significantly enhances model performance, especially in Emotion and Hate Speech detection, emphasising the significance of contextualised learning. In addition, larger multilingual models like Qwen-2.5-72B and Gemma-2-27B are uniformly superior on all tasks, demonstrating the joint benefits of model scale and high-quality multilingual pretraining.

Small models Mistral-V3-7B, Phi-4-14B, and DeepSeek-R1-8B perform relatively less well in general on the all tasks, especially under zero-shot. Their relatively low performance might result from a smaller number of parameters and the amount of languages they were less exposed to when being pretrained. Models pre-trained on English (or other high-resource languages) often overlook unimodal and linguistic conditions, missing out because of the absence of coverage for low-resource languages, as well as less-than-desirable syntactic or cultural alignment. This highlights the importance of scale and multilingual data diversity for generalisation across low-resource languages. Successful performance of Qwen-2.5-72B and Gemma-2-27B is thus a testament to not only their higher capacity models but also the amount of pretraining they have received multilingually, enabling greater adaptability to morphologically rich languages like Bengali.

Differences in model behaviour are also evident at the task level. Sentiment analysis is relatively simple yet only achieves moderate results, i.e. best models obtain 62–63% F1 scores. The conundrum is how to distinguish neutral orientation from the negative one, as could be seen in the confusion matrix result. The emotion recognition still remains the most challenging task, while the state-of-the-art method yields 65.76% F1-score in few-shot scenario. This limitation might be due to the fine-grained nature of our task where we have to discriminate between six very similar feelings that often share a portion of appearances in Bengali.

Results of hate speech detection are more promising, with state-of-the-art approaches approaching 80% F1-score. The binary nature of the task, and better linguistic cues for hate speech make it relatively easier. However, the error analysis also indicates a small over-prediction as compared to the hateful labels that may imply a conservative classification

approach, good for moderation, but prone to raising false positives. Fake News detection is the most tractable task, with F1-scores over 84% across different models. This high accuracy is most likely due to the clear stylistic and factual differences between real and fake articles.

Few-shot learning effect is task-dependent. It proves effective for sentiment analysis, hate speech detection, and emotion recognition consistently, while its impact is constrained or slightly negative regarding fake news detection. This tendency indicates that fake news detection benefits mostly from previous knowledge of facts and pretrained representations, rather than from abundant in-context examples. On the other hand, ones requiring fine-grained tones or affects such as sentiment or emotion classification benefit significantly from just a few labelled examples; this provides strong evidence for the importance of in-context learning in linguistic subtlety under low-resource settings.

In addition to these discoveries, the analysis yields other overarching conclusions. Recognizing emotions is still the hardest thing to do because many Bengali ways of showing emotions are quite peculiar to their culture or situation. In real conversation, the lines between feelings like anger, disgust, and melancholy are typically very thin, which makes it harder for models to tell them apart without clear instances. On the other hand, Qwen-2.5-72B and Gemma-2-27B always work well since they have superior multilingual pretraining, cover more tokens, and represent Indic scripts better. These traits let them pick up on morphological changes, compound forms, and subtle semantic differences that smaller models don't always catch. More broadly, the results show that there are some important rules for low-resource NLP: model scale alone isn't enough; it's important to have a variety of multilingual data; in-context examples are especially useful for tasks that involve fine-grained linguistic signals; and balanced evaluation datasets are important for fair comparison across LLMs. These results show that Bengali NLP will only get better in the future if there are more language-specific resources and better multilingual training pipelines.

A. IMPLICATIONS FOR LOW-RESOURCE LANGUAGE NLP

Findings from this research study carry several crucial results for furthering NLP in languages with low-resources like Bengali. First, the wide gap in performance between large-scale models (Qwen-2.5-72B and Gemma-2-27B) and small ones (LLaMA-3.2-3B and Mistral-V3-7B) proves that large model sizes continue to play a decisive role in dealing with the arithmetic diversity and morphological complexity of languages. But this also results in an accessibility challenge, as it remains a significant obstacle for many researchers and institutions working in resource-constrained environments the computational and hardware requirements of big models.

Second, the consistent effectiveness of few-shot prompting

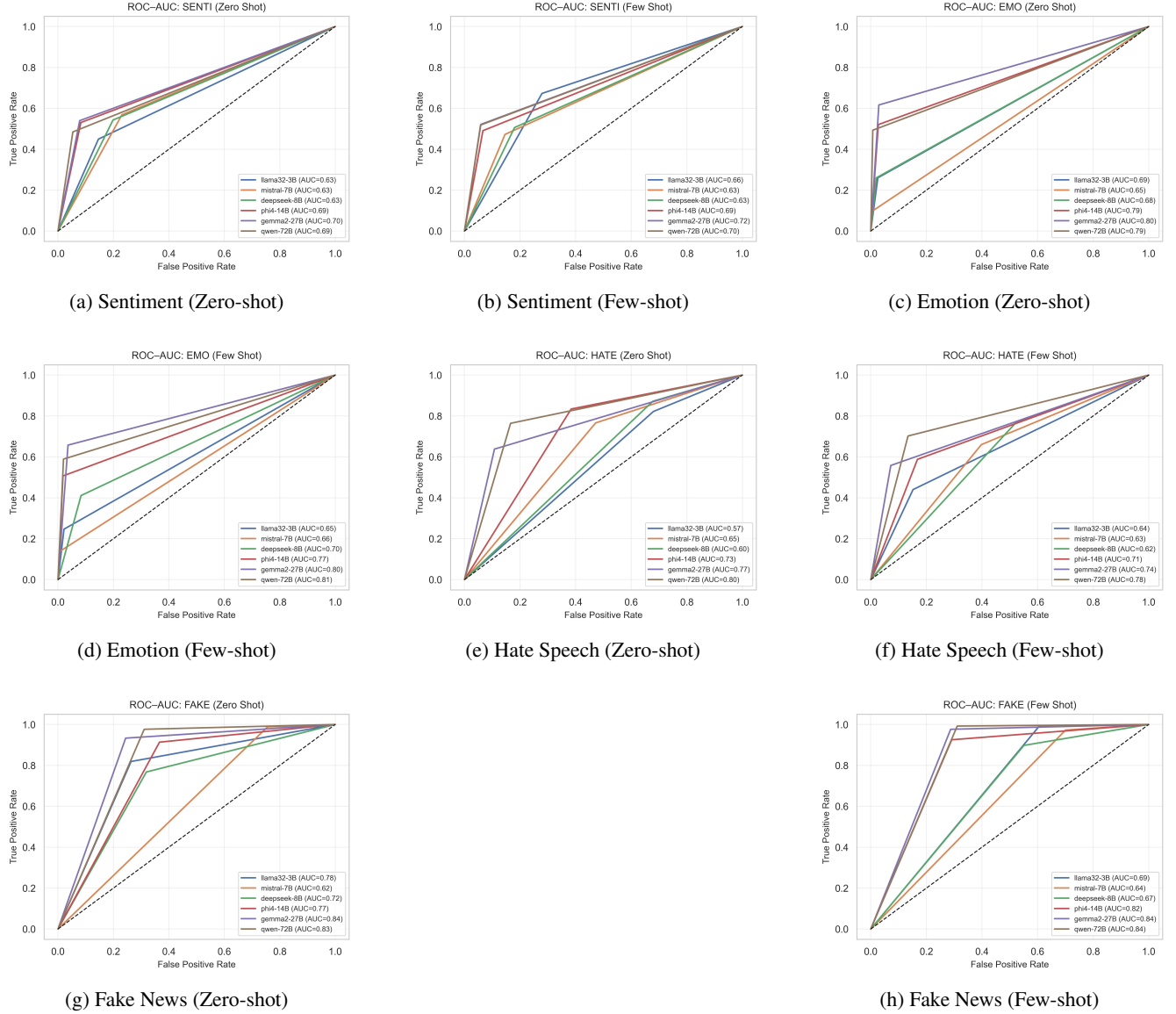


FIGURE 17: ROC-AUC curves of six LLMs on Bengali text classification tasks under zero- and few-shot settings.

on most tasks shows that meaningful performance gains can still be achieved with limited labeled data. This finding is particularly valuable for languages without large annotated datasets, where creating large corpora is difficult or expensive. With the creation of small but high-quality labelled datasets, future research in Bengali and other similar languages should move away from only looking at developing huge corpora. These datasets are well-suited to few-shot learning, allowing LLMs to infer effectively with few examples whilst keeping computational demands limited.

Thirdly, differences in task difficulty demonstrate that the choice of method and problem framing significantly affect model performance. Tasks with clear-cut categorical distinctions—e.g., hate speech and fake news detection can be quite directly solved using our existing LLM

architectures. In contrast, fine-grained tasks like emotion recognition, which require thinking deeply about subtle expressions and environmental cues, might profit from different methods such as fine-tuning, ensemble learning or hybrids of LLMs with task-specific classifiers. The development of adaptive methods that balance generalisation against linguistic sensitivity could make models more robust in complex low-resource language environments.

B. LIMITATIONS

While this work offers some insights regarding their performance towards a few of the available open-source LLMs for Bangla text classification, it has its own limitations, which can be defined as follows. Details of the limitation of the work In this work, the evaluation is restricted to classification tasks, which is a significant subset of NLP,

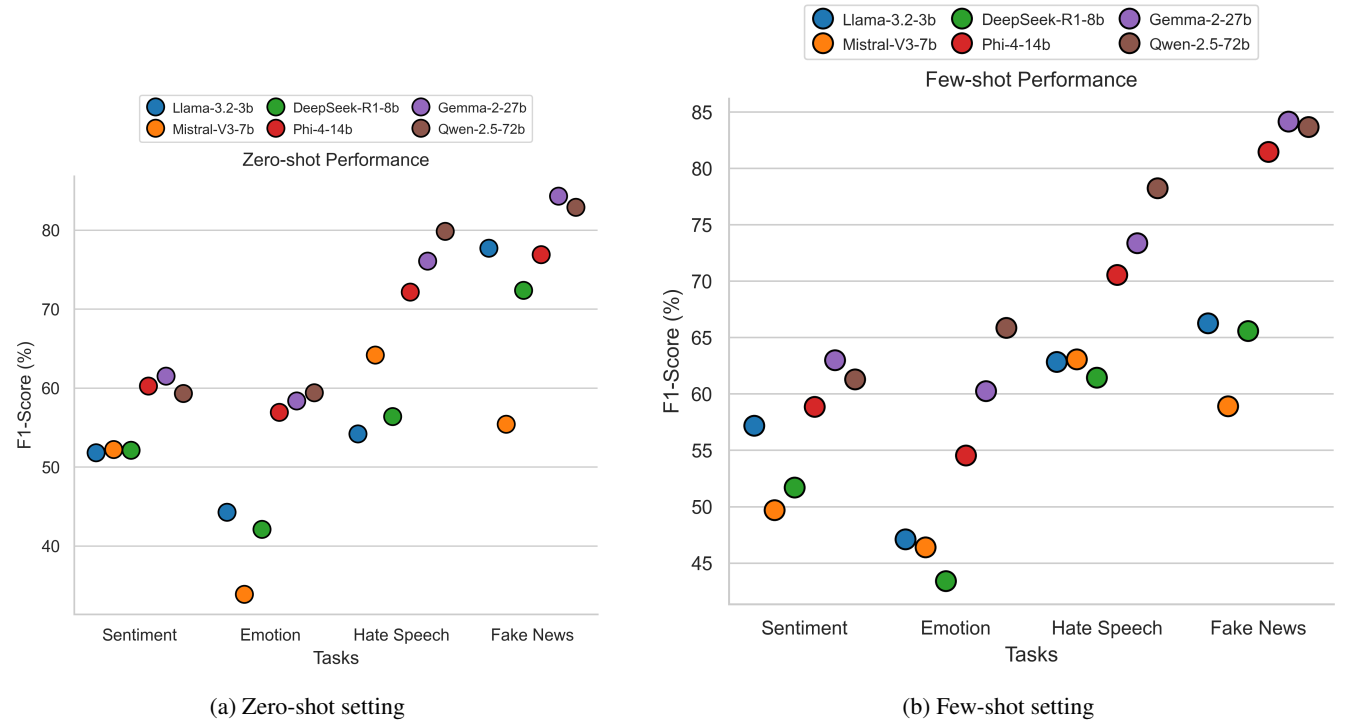


FIGURE 18: LLMs' performance with two prompt settings across various classification tasks.

side by side with many other directions like text generation, summarisation, or question answering in Bengali. Future developments that leverage this analysis on more tasks for a broader range of linguistic phenomena will allow for more general conclusions about LLM performance in more nuanced linguistic contexts.

In addition, for easier transparency and reproducibility, the research will only use publicly available models. However, this selection excludes the comparison to closed-source proprietary systems (such as GPT-4 or Claude), which might have a better overall performance. They should make it to future benchmarks — providing a better perspective of how wide the open-closed ecosystem gap is for Bengali language processors.

Third, the prompting method adopted here is exclusive to zero-shot and few-shot (five-shot) settings. More advanced methods of prompting (e.g., chain-of-thought reasoning, self-consistency or retrieval-augmented generation) could lead to more improvements but were not in the scope of this work. Also, parameter-efficient fine-tuning mechanisms such as LoRA or QLoRA, could be pursued and provide practical ways to improve the performance of smaller models without spending much in terms of computing time.

Fourth, the datasets we used, while standard benchmarks in the Bengali NLP community, suffer from natural limitations in terms of size and coverage, domain diversity, and annotation consistency. Joboji says that much of their data comes from social media, and may not reflect formal or domain-specific Bengali. Second, some datasets

are small (e.g., the emotion recognition corpus: only 625 examples), which may compromise statistical stability and the generalisation value of the results.

Finally, although we focus on quantitative performance measures (i.e., accuracy or F1-score) in this work, we do not delve into more holistic notions of fairness, bias, and cultural relevance of model predictions. Because of the sensitive nature of tasks like hate speech detection and fake news classification, future work must also include bias assessment frameworks to analyse fine-grained model behaviour in different demographic groups, social context, and dialectic variation within the Bengali-speaking community.

VI. CONCLUSION

A comprehensive evaluation of six open-source Large Language Models (LLMs)—LLaMA-3.2-3B, Mistral-V3-7B, DeepSeek-R1-8B, Phi-4-14B, Gemma-2-27B, and Qwen-2.5-72B is presented with this research, where four Bengali text classification tasks are evaluated, such as sentiment analysis, emotion recognition, hate-speech detection, and fake-news detection. For ensuring equal and consistent comparisons, both zero-shot and few-shot prompting strategies were applied along with vLLM. The outcome of this research is that it sheds light on the fact that the larger multilingual models constantly surpass smaller ones across all tasks. With zero-shot setting, the outperformed model is Qwen-2.5-72B with 79.9% accuracy and an F1-score of 79.88%, whereas for fake-news classification, 84.25% accuracy is achieved with

Gemma-2-27B as well as 84.33% F1-score. After a few-shot prompt was applied, Gemma-2-27B maintained strong performance with an F1-score of 84.17%. Qwen-2.5-72B achieved the best emotion-recognition result, reaching an F1-score of 65.86%. These results suggest that the size of the model, the quality of multilingual pretraining, and the use of in-context examples all play an important role in improving performance for low-resource languages such as Bengali.

A. FUTURE WORK

Though the findings are pretty favourable, several issues still need careful attention. A difficulty is seen among the models that are working to classify the neutral as well as the negative statement, and they also frequently lump together emotions with similar meanings, such as anger and disgust, as well as misleading statements, which are also labelled as real. These mechanisms can be strengthened further in the future. For improving the ability of the model, approaches that are improved should be applied, such as step-by-step reasoning or the use of external information sources. With an intention to improve the performance, along with the requirement of not using expensive hardware, Lightweight fine-tuning approaches such as LoRA or QLoRA can be applied. In order to acquire an improved awareness of their overall capabilities and limits, a broader set of NLP tasks, for instance, text generation, summarisation, and question answering, should be tested as well. Additionally, retaining the NLP system, which is a blessing for Bangali users, should pay great attention to its fairness across different social and demographic groups. With a view to improving further, larger and more diverse datasets should be introduced, which could be a great help to Bangali NLP tool users.

Data Availability Statement: The datasets used in this study are publicly available at: <https://github.com/hamidhosen42/open-source-llms-bengali-text-classification>.

REFERENCES

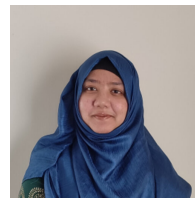
- [1] H. Dehan, T. Rahman, A. S. Kabir, and M. Hasan, "TinyLLM efficacy in low-resource language: An experiment on Bangla text classification," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (ACL)*, Bangkok, Thailand, 2024, pp. 245–258.
- [2] K. M. A. Rahman, F. B. Kabir, M. A. Rusho, S. Ahmed, and T. Islam, "Bengali sentiment analysis: Fine-tuning LLMs and comparing with traditional ML models for Bengali movie reviews," in *Proc. Int. Conf. Electr. Comput. Energy Technol. (ICECET)*, Cape Town, South Africa, 2023, pp. 1–6.
- [3] F. Kabir, S. I. Khan, and A. S. Rahman, "BenLLM-Eval: A comprehensive evaluation into the potentials and pitfalls of large language models on Bengali NLP," *arXiv preprint arXiv:2309.13173*, 2023.
- [4] A. Bhattacharjee, T. Hasan, K. Ahmad, M. S. Kanan, A. Basak, M. S. H. Shahriyar, M. F. Alam, F. M. J. Mehedi Hasan, I. H. Sarker, and R. Amin, "BanglaBERT: Language model pretraining and benchmarks for low-resource language understanding evaluation in Bangla," in *Findings Assoc. Comput. Linguistics: NAACL*, Online, 2022, pp. 1318–1327.
- [5] H. Mahmud and R. Mahmud, "Enhancing sentiment analysis in Bengali texts: A hybrid approach using lexicon-based algorithm and pretrained language model BanglaBERT," *IEEE Access*, vol. 12, pp. 145672–145683, 2024.

- [6] S. Bhowmik, T. T. Dipto, M. S. Islam, S. Hsu, and T. Reasat, "Evaluating LLMs' multilingual capabilities for Bengali: Benchmark creation and performance analysis," *arXiv preprint arXiv:2507.23248*, 2025.
- [7] Z. A. Nazi, M. R. Hossain, and F. A. Mamun, "Evaluation of open and closed-source LLMs for a low-resource language with zero-shot, few-shot, and chain-of-thought prompting," *Natural Language Processing J.*, vol. 10, p. 100124, 2025.
- [8] R. Prome, F. T. J. Faria, and M. S. Rahman, "Leveraging the potential of prompt engineering for hate speech detection in low-resource languages," *IEEE Access*, vol. 12, pp. 89765–89777, 2024.
- [9] B. Banda, N. Vashisth, and S. Banerjee, "A few-shot learning approach for a multilingual agro-information question answering system," *Comput. Intell. Neurosci.*, vol. 2024, Article ID 5562073, 2024.
- [10] A. Raihan and M. Zampieri, "TigerLLM: A family of Bangla large language models," *arXiv preprint arXiv:2501.04527*, 2025.
- [11] T. Mahfuz, S. I. Chowdhury, and F. N. Ahmed, "Too late to train, too early to use: A study on necessity and viability of low-resource Bengali LLMs," *Findings Assoc. Comput. Linguistics: EMNLP*, Singapore, 2024, pp. 2978–2989.
- [12] M. Mahtab, A. Rahman, and F. Islam, "BanNERD: A benchmark dataset and context-driven approach for Bangla named entity recognition," *IEEE Access*, vol. 12, pp. 110297–110309, 2024.
- [13] S. Shafayat, M. S. Islam, and A. Rahman, "BenQA: A question answering benchmark for Bengali and English," *Proc. Int. Conf. Comput. Linguistics (COLING)*, Torino, Italy, 2024, pp. 215–226.
- [14] A. Singh, R. Mishra, and A. Choudhury, "IndicGenBench: A multilingual benchmark to evaluate generation," *arXiv preprint arXiv:2403.18356*, 2024.
- [15] S. Shukla, P. Bhattacharya, and S. Dey, "IndicMMLU-Pro: Benchmarking Indic large language models on multi-task language understanding," *arXiv preprint arXiv:2501.12791*, 2025.
- [16] M. A. Fahim, A. H. Rahman, and A. Islam, "BanglaTLit: A benchmark dataset for back-transliteration of Romanized Bangla," *Proc. LREC-COLING 2024*, Torino, Italy, 2024, pp. 1788–1799.
- [17] N. Haider, S. M. Rahman, and T. Hasan, "BANTH: A multi-label hate speech detection dataset for transliterated Bangla," *IEEE Access*, vol. 12, pp. 107642–107653, 2024.
- [18] R. Romim, S. Ahmed, and K. Ali, "BD-SHS: A benchmark dataset for learning to detect online Bangla hate speech in different social contexts," *IEEE Access*, vol. 12, pp. 103213–103225, 2024.
- [19] M. Nayeem, A. S. Chowdhury, and F. Alam, "MultiBanFakeDetect: Integrating advanced fusion techniques for multimodal detection of Bangla fake news in under-resourced contexts," *Inf. Process. Manage.*, vol. 62, no. 4, p. 103672, 2025.
- [20] M. Hasan, M. A. Khan, and S. S. Chowdhury, "Aspect-based sentiment analysis datasets for Bangla text," *Data Brief*, vol. 52, p. 110687, 2025.
- [21] S. K. Nahin, R. N. Nandi, S. Sarker, Q. S. Muhtaseem, M. Kowsher, A. C. Shill, M. I. Menon, T. A. Muntasir, and F. Alam, "TituLLMs: A family of Bangla LLMs with comprehensive benchmarking," in *Findings Assoc. Comput. Linguistics: ACL*, Bangkok, Thailand, 2025, pp. 24922–24940.
- [22] F. T. J. Faria, L. H. Baniata, and S. Kang, "Investigating the predominance of large language models in low-resource Bangla language over transformer models for hate speech detection: A comparative analysis," *Mathematics*, vol. 12, no. 3687, pp. 1–40, 2024.
- [23] M. A. Hasan, S. Das, A. Anjum, F. Alam, and S. R. H. Noori, "Zero-and few-shot prompting with LLMs: A comparative study with fine-tuned models for Bangla sentiment analysis," in *Proc. LREC-COLING*, Torino, Italy, 2024, pp. 17808–17818.
- [24] S. A. Rafid, F. B. Sakib, and M. S. Rahman, "Zero-shot multi-label classification of Bangla documents: Large decoders vs. classic encoders," *arXiv preprint arXiv:2502.08753*, 2025.
- [25] R. Saha, A. H. Chowdhury, and F. H. Shah, "Automatic detection of manipulated Bangla news: A new knowledge-driven approach," *IEEE Access*, vol. 13, pp. 119876–119888, 2025.
- [26] T. Hossain, A.-R. Islam, M. H. K. Mehedi, A. A. Rasel, M. Abdullah-AL-Wadud, and J. Uddin, "BanglaNewsClassifier: A machine learning approach for news classification in Bangla newspapers using hybrid stacking classifiers," *PLOS ONE*, vol. 20, no. 6, p. e0321291, 2025.
- [27] A. K. Chowdhury, S. R. Sujon, M. S. S. Shafi, T. Ahmmad, S. Ahmed, K. M. Hasib, and F. M. Shah, "Harnessing large language models over transformer models for detecting Bengali depressive social media text: A comprehensive study," *Natural Language Processing J.*, vol. 7, p. 100075, 2024.

- [28] M. Kamruzzaman, M. M. I. Shovon, and G. L. Kim, "BanMANI: A dataset to identify manipulated social media news in Bangla," *arXiv preprint arXiv:2311.02570*, 2023.
- [29] F. T. J. Faria, M. B. Moin, A. I. Fahim, P. Debnath, and F. M. Shah, "Unraveling the dominance of large language models over transformer models for Bangla natural language inference: A comprehensive study," *arXiv preprint arXiv:2405.02937*, 2024.
- [30] S. Maity and A. Deroy, "Leveraging prompt-tuning for Bengali grammatical error explanation using large language models," *arXiv preprint arXiv:2504.05642*, 2025.
- [31] K. I. Islam, S. Kar, M. S. Islam, and M. R. Amin, "SentNoB: A dataset for analysing sentiment on noisy Bangla texts," in *Findings Assoc. Comput. Linguistics: EMNLP*, Punta Cana, Dominican Republic, Nov. 2021, pp. 3265–3271.
- [32] A. Das, O. Sharif, M. M. Hoque, and I. H. Sarker, "Emotion classification in a resource constrained language using transformer-based approach," in *Proc. Conf. North American Chapter Assoc. Comput. Linguistics: Student Research Workshop (NAACL-SRW)*, Online, Jun. 2021, pp. 150–158.
- [33] M. R. Karim, S. K. Dey, T. Islam, S. Sarker, M. H. Menon, K. Hossain, M. A. Hossain, and S. Decker, "DeepHateExplainer: Explainable hate speech detection in under-resourced Bengali language," in *Proc. IEEE 8th Int. Conf. Data Sci. Adv. Analytics (DSAA)*, Porto, Portugal, Oct. 2021, pp. 1–10.
- [34] M. Z. Hossain, M. A. Rahman, M. S. Islam, and S. Kar, "BanFakeNews: A dataset for detecting fake news in Bangla," *arXiv preprint arXiv:2004.08789*, Apr. 2020.
- [35] T. Brown, B. Mann, N. Ryder *et al.*, "Language Models Are Few-Shot Learners," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, Vancouver, Canada, Dec. 2020, pp. 1877–1901.
- [36] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, E. Grave, and G. Lample, "LLaMA: Open and efficient foundation language models," *arXiv preprint arXiv:2302.13971*, Feb. 2023. [Online]. Available: <https://arxiv.org/abs/2302.13971>
- [37] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier, L. R. Lavaud, M.-A. Lachaux, P. Stock, T. Le Scao, T. Lavril, T. Wang, T. Lacroix, and W. El Sayed, "Mistral 7B," *arXiv preprint arXiv:2310.06825*, Oct. 2023. [Online]. Available: <https://arxiv.org/abs/2310.06825>
- [38] DeepSeek-AI, D. Guo, D. Yang, H. Zhang, J. Song, R. Zhang, R. Xu, *et al.*, "DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning," *arXiv preprint arXiv:2501.12948*, Jan. 2025. [Online]. Available: <https://arxiv.org/abs/2501.12948>
- [39] M. Abidin, J. Aneja, H. Behl, S. Bubeck, R. Eldan, S. Gunasekar, M. Harrison, R. J. Hewett, M. Javaheripi, P. Kauffmann, *et al.*, "Phi-4 Technical Report," *arXiv preprint arXiv:2412.08905*, Dec. 2024. [Online]. Available: <https://arxiv.org/abs/2412.08905>
- [40] Gemma Team, M. Riviere, S. Pathak, P. G. Sessa, C. Hardin, S. Bhupatiraju, L. Hussenot, T. Mesnard, B. Shahriari, A. Ramé, *et al.*, "Gemma 2: Improving Open Language Models at a Practical Size," *arXiv preprint arXiv:2408.00118*, Aug. 2024. [Online]. Available: <https://arxiv.org/abs/2408.00118>
- [41] A. Yang, B. Yang, B. Zhang, B. Hui, B. Yu, C. Gao, C. Huang, C. Lv, F. Huang, J. Zhou, *et al.*, "Qwen2.5 Technical Report," *arXiv preprint arXiv:2412.15115*, Dec. 2024. [Online]. Available: <https://arxiv.org/abs/2412.15115>
- [42] J. Ainslie, J. Lee-Thorp, M. De Jong, Y. Zemlyanskiy, F. Lebrón, and S. Sanghai, "GQA: Training generalized multi-query transformer models from multi-head checkpoints," *arXiv preprint arXiv:2305.13245*, May 2023.
- [43] S. Cahyawijaya, H. Lovenia, and P. Fung, "LLMs are few-shot in-context low-resource language learners," *arXiv preprint arXiv:2403.16512*, Mar. 2024.
- [44] D. M. W. Powers, "Evaluation: from precision, recall and F-measure to ROC, informedness, markedness and correlation," *arXiv preprint arXiv:2010.16061*, Oct. 2020. [Online]. Available: <https://arxiv.org/abs/2010.16061>
- [45] M. R. Karim, B. R. Chakravarthi, J. P. McCrae, and M. Cochez, "Classification benchmarks for under-resourced Bengali language based on multichannel convolutional-LSTM network," in *Proc. IEEE 7th Int. Conf. Data Sci. Adv. Analytics (DSAA)*, Sydney, NSW, Australia, 2020, pp. 390–399.
- [46] G. S. Kohli, A. Sen, S. Sekhar, S. Sahoo, S. Parida, S. R. Dash, and O. Bojar, "BengaliGPT: An instruction following LLaMA model for Bengali,"
- [47] J. Sikder, P. Chakraborty, U. K. Das, and K. Dhar, "A hybrid approach for Bengali sentence validation," *Artif. Intell. Rev.*, vol. 57, no. 11, p. 316, 2024.
- [48] A. K. Zehady, S. A. Mamun, N. Islam, and S. Karmaker, "BongLLaMA: LLaMA for Bangla language," *arXiv preprint arXiv:2410.21200*, 2024. [Online]. Available: <https://arxiv.org/abs/2410.21200>



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